

zTool Interface

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Due to limited time, development of this package is currently suspended.

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1 Basic Introduction

The $\text{\texttt{zT\TeX}}$ macro collection has independently implemented a `ztool` package, which contains all the commands from the now-obsolete `l3sys-shell` package. In addition, `ztool` provides functions related to box operations, file IO, and basic graphics drawing. With the help of `ztool`, $\text{\texttt{zT\TeX}}$ can avoid or reduce the use of the `-shell-escape` command-line parameter or calls to other related packages (such as the `robust-externalize` package).

The GitHub address of this package is:

https://github.com/zongpingding/zTeX_bundle

This repository contains the source code and user manual of this macro collection. The stable version of the current macro collection was released in September 2025. For the latest development version, please switch to the “dev” branch. This manual applies to the current latest development version.

2 Package Options

ztool is divided into four libraries: “shell-escape, file-io, box, zdraw”. Each library is independent and can be loaded separately. By default, ztool does not load any library.

ztool/shell-escape	shell-escape = <code><false true></code> Initial value: <code>false</code>
ztool/file-io	file-io = <code><false true></code> Initial value: <code>false</code>
ztool/box	box = <code><false true></code> Initial value: <code>false</code>
ztool/zdraw	zdraw = <code><false true></code> Initial value: <code>false</code>

New: 2025-05-22

These four options are options for the ztool package and can be used when loading the ztool package. A basic usage example is as follows, which loads the shell-escape and box libraries of ztool:

<code>\usepackage[shell-escape, box=true]{ztool}</code>	Ex. 1
---	--------------

<code>\ztoolloadlib</code>	<code>\ztoolloadlib {<library>}</code>
----------------------------	--

New: 2025-05-22

This command is used to load a ztool library. `<library>` is the name of the library, and the optional values are: “shell-escape, file-io, box, zdraw”.

A basic usage example is as follows, which loads the shell-escape and box libraries of ztool:

<code>\ztoolloadlib{shell-escape, box}</code>	Ex. 2
---	--------------

3 l3sys-shell

This section mainly introduces the commands from the original l3sys-shell package implemented in ztool. Therefore, when using the commands in this section, you need to enable the `-shell-escape` parameter when compiling the L^AT_EX document, otherwise this series of commands will not perform any operations.

WARNING: Please use the commands in this section with caution. Some improper operations may lead to irreparable consequences.

```
\ztool_shell_escape:n
\ztool_shell_escape:e
```

Updated: 2024-12-05

```
\ztool_shell_escape:n {⟨command⟩}
```

When the `-shell-escape` parameter is enabled, this command will execute `⟨command⟩` in the shell. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_mkdir:n
\ztool_shell_mkdir:e
```

Updated: 2024-12-05

```
\ztool_shell_mkdir:n {⟨dir⟩}
```

When the `-shell-escape` parameter is enabled, this command will create a directory `⟨dir⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_cp:nn
\ztool_shell_cp:(ee|ne|en)
```

Updated: 2024-12-05

```
\ztool_shell_cp:nn {⟨source⟩}{⟨target⟩}
```

When the `-shell-escape` parameter is enabled, this command will copy the file `⟨source⟩` to the file `⟨target⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_mv:nn
\ztool_shell_mv:(ee|ne|en)
```

Updated: 2024-12-05

```
\ztool_shell_mv:nn {⟨source⟩}{⟨target⟩}
```

When the `-shell-escape` parameter is enabled, this command will move the file `⟨source⟩` to the directory `⟨target⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_rm:n
\ztool_shell_rm:e
```

Updated: 2024-12-05

```
\ztool_shell_rm:n {⟨file⟩}
```

When the `-shell-escape` parameter is enabled, this command will delete the file `⟨file⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_rmdir:n
\ztool_shell_rmdir:e
```

Updated: 2024-12-05

```
\ztool_shell_rmdir:n {⟨dir⟩}
```

When the `-shell-escape` parameter is enabled, this command will delete the directory `⟨dir⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_get_shell_pwd:N
\ztool_get_shell_pwd:c
```

Updated: 2024-12-05

```
\ztool_get_shell_pwd:N ⟨tl⟩
```

When the `-shell-escape` parameter is enabled, this command will return the current working directory and store it in `⟨tl⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

`\ztool_shell_split_ls:nN`

`Updated: 2024-12-05`

`\ztool_shell_split_ls:nN {<dir>}<t1>`

When the `-shell-escape` parameter is enabled, this command will return all filenames under the directory `<dir>` and store them in `<t1>`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

4 File IO

This section mainly introduces the file IO operations implemented in `ztool`, including: reading files, writing files, appending to files, etc. This series of commands does not require enabling the `-shell-escape` parameter.

<code>\ztool_file_new:nn</code>	<code>\ztool_file_new:nn {<bool>}{<file>}</code>
---------------------------------	--

Updated: 2024-12-05

This command is used to create a new file named `<file>`. If `<file>` does not exist, a new file named `<file>` will be created. If the file already exists, then when `<bool>` is `\c_true_bool`, **it will overwrite the original file**, otherwise no operation will be performed.

<code>\ztool_read_file_keep_spaces:nn</code>	<code>\ztool_read_file_keep_spaces:nn {<file>}{<code>}</code>
--	---

New: 2025-09-12

This command is similar to the `\ior_str_map_inline:Nn` command in `l3file`, but `\ztool_read_file_keep_spaces:nn` will change the catcode of the content to `\c_document_cctab`. In `<code>`, the variable `\l__ztool_strmap_read_tl` represents the content of each line. **Note:** `<file>` needs to include the file extension; if `<file>` does not exist, this command will not perform any operation.

<code>\ztool_read_file_as_seq:nnN</code>	<code>\ztool_read_file_as_seq:nnN {<bool>}{<file>}{<seq>}</code>
<code>\ztool_read_file_as_seq:(neN nnc nec)</code>	

Updated: 2024-12-05

This command is used to read the content of the file `<file>` and store it in `<seq>`. If `<file>` does not exist, `<seq>` will be set to empty. `<bool>` is used to control whether to keep the spaces at the **end of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if `<bool>` is `\c_true_bool`, then the spaces at the **end of the line** are kept, otherwise they are not.

NOTE:

1. The definition of `<seq>` is local;
2. Due to the limitation of the command `\ior_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the beginning of the line;
3. The catcode of the content in `<seq>` is the current catcode.

<code>\ztool_read_file_as_seq_keep_spaces:nnN</code>	<code>\ztool_read_file_as_seq_keep_spaces:nnN</code>
<code>\ztool_read_file_as_seq_keep_spaces:(neN nnc nec)</code>	<code>{⟨bool⟩}{⟨file⟩}{⟨seq⟩}</code>

New: 2025-09-01

This command is used to read the content of the file $\langle file \rangle$, **preserving internal spaces**, and store it in $\langle seq \rangle$. If $\langle file \rangle$ does not exist, $\langle seq \rangle$ will be set to empty. $\langle bool \rangle$ is used to control whether to keep the spaces at the **beginning of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if $\langle bool \rangle$ is `\c_true_bool`, then the spaces at the **beginning of the line** are kept, otherwise they are not.

NOTE:

1. The definition of $\langle seq \rangle$ is local;
2. Due to the limitation of the command `\ior_str_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the end of the line;
3. The catcode of the content in $\langle seq \rangle$ is modified to `\c_document_cctab`.

<code>\ztool_gread_file_as_seq:nnN</code>	<code>\ztool_read_file_as_seq:nnN {⟨bool⟩}{⟨file⟩}{⟨seq⟩}</code>
<code>\ztool_gread_file_as_seq:(neN nnc nec)</code>	

Updated: 2025-01-05

This command is used to read the content of the file $\langle file \rangle$ and store it in $\langle seq \rangle$. If $\langle file \rangle$ does not exist, $\langle seq \rangle$ will be set to empty. $\langle bool \rangle$ is used to control whether to keep the spaces at the **end of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if $\langle bool \rangle$ is `\c_true_bool`, then the spaces at the **end of the line** are kept, otherwise they are not.

NOTE:

1. The definition of $\langle seq \rangle$ is global;
2. Due to the limitation of the command `\ior_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the beginning of the line;
3. The catcode of the content in $\langle seq \rangle$ is the current catcode.

<code>\ztool_gread_file_as_seq_keep_spaces:nnN</code>	<code>\ztool_gread_file_as_seq_keep_spaces:nnN</code>
<code>\ztool_gread_file_as_seq_keep_spaces:(neN nnc nec)</code>	<code>{⟨bool⟩}{⟨file⟩}{⟨seq⟩}</code>

New: 2025-09-01

This command is used to read the content of the file `⟨file⟩`, **preserving internal spaces**, and store it in `⟨seq⟩`. If `⟨file⟩` does not exist, `⟨seq⟩` will be set to empty. `⟨bool⟩` is used to control whether to keep the spaces at the **beginning of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if `⟨bool⟩` is `\c_true_bool`, then the spaces at the **beginning of the line** are kept, otherwise they are not.

NOTE:

1. The definition of `⟨seq⟩` is global;
2. Due to the limitation of the command `\ior_str_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the end of the line;
3. The catcode of the content in `⟨seq⟩` is modified to `\c_document_cctab`.

<code>\ztool_write_seq_to_file:nNn</code>	<code>\ztool_write_seq_to_file:nNn {⟨bool⟩}{⟨seq⟩}{⟨file⟩}</code>
<code>\ztool_write_seq_to_file:(nNe nNV nce ncV)</code>	

New: 2025-05-27

This command is used to write `⟨seq⟩` line by line to the file `⟨file⟩`. If `⟨file⟩` does not exist, a new file named `⟨file⟩` will be created; if `⟨file⟩` already exists, you can use `⟨bool⟩` to control the current writing mode: when `⟨bool⟩` is `\c_true_bool`, **overwrite**; when `⟨bool⟩` is `\c_false_bool`, **append**; if `⟨seq⟩` is empty, no operation will be performed.

<code>\ztool_append_to_file:nn</code>	<code>\ztool_append_to_file:nn {⟨file⟩}{⟨content⟩}</code>
<code>\ztool_append_to_file:(no nf ee)</code>	

Updated: 2025-01-05

This command is used to append `⟨content⟩` to the file `⟨file⟩`. If `⟨file⟩` does not exist, a new file named `⟨file⟩` will be created and `⟨content⟩` will be written into it.

<code>\ztool_replace_file_line:nnn</code>	<code>\ztool_replace_file_line:nnn {⟨file⟩}{⟨line⟩}{⟨content⟩}</code>
<code>\ztool_replace_file_line:(enn ene eee)</code>	

Updated: 2025-01-05

This command is used to replace the `⟨line⟩`-th line in the file `⟨file⟩` with `⟨content⟩`. If `⟨file⟩` does not exist, no operation will be performed.

<code>\ztool_insert_to_file:nnn</code>	<code>\ztool_insert_to_file:nnn {⟨file⟩}{⟨line⟩}{⟨content⟩}</code>
<code>\ztool_insert_to_file:(nen nfn een)</code>	

Updated: 2025-01-05

This command is used to insert `⟨content⟩` before the `⟨line⟩`-th line of the file `⟨file⟩`. If `⟨file⟩` does not exist, no operation will be performed.

The following example shows how to use several file IO operation commands in `ztool`:

Ex. 3

```
% \usepackage{verbatim}
\ExplSyntaxOn
\ztool_file_new:nn {\c_true_bool}{testIO.txt}
\seq_new:N \l_ztool_tmp_seq \seq_clear:N \l_ztool_tmp_seq
\ztool_append_to_file:nn {testIO.txt} {|APPEND-CONTENT|}
\ztool_insert_to_file:nnn {testIO.txt} {1} {|INSERT-~-CONTENT|}
\ztool_append_to_file:nn {testIO.txt} {|APPEND-CONTENT-II|}
\ztool_replace_file_line:nnn {testIO.txt} {3} {|REPLACE-CONTENT|}
\ztool_gread_file_as_seq:nnN {\c_false_bool} {testIO.txt}
\l_ztool_tmp_seq
\seq_use:Nn \l_ztool_tmp_seq {\par}
\ExplSyntaxOff
\verbatiminput*{testIO.txt}
```

```
|INSERT-CONTENT|
|APPEND-CONTENT|
|REPLACE-CONTENT|

|INSERT-~-CONTENT|
|APPEND-CONTENT|
|REPLACE-CONTENT|
```

The following example demonstrates how the file to seq series of commands in `ztool` handle spaces. The content in the file `testSpaces.txt` is as follows:

```
AAAA
BB_ _BB
_ _CC_ _CC_ _
DDDD
```

Ex. 4

```
\ExplSyntaxOn
\def\TTTa#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq:nnN {\c_true_bool} {#1} \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}
```

```

\def\TTTb#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq:nnN {\c_false_bool} {#1} \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}

\def\TTTc#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq_keep_spaces:nnN {\c_true_bool} {#1} ↙
  \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}

\def\TTTd#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq_keep_spaces:nnN {\c_false_bool} {#1} ↙
  \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}

\TTTa{testSpaces.txt}
\TTTb{testSpaces.txt}
\TTTc{testSpaces.txt}
\TTTd{testSpaces.txt}
\ExplSyntaxOff
\TTTa{testSpaces.txt}
\TTTb{testSpaces.txt}
\TTTc{testSpaces.txt}
\TTTd{testSpaces.txt}

```

Results displayed in the command line in various cases (somewhat simplified):

\TTTa: outside expl3, true

```

{AAAA_}
{BB_BB_}
{CC_CC_}
{DDDD_}.

```

\TTTb: outside expl3, false

```

{AAAA}
{BB_BB}
{CC_CC}
{DDDD}.

```

\TTTa: inside expl3, true

{AAAA}
{BBBB}
{CCCC}
{DDDD}.

\TTTb: inside expl3, false

{AAAA}
{BBBB}
{CCCC}
{DDDD}.

.....

\TTTc: outside expl3, true

{AAAA}
{BB_BB}
{_CC_CC}
{DDDD}.

\TTTd: outside expl3, false

{AAAA}
{BB_BB}
{CC_CC}
{DDDD}.

\TTTc: inside expl3, true

{AAAA}
{BB_BB}
{_CC_CC}
{DDDD}.

\TTTd: inside expl3, false

{AAAA}
{BB_BB}
{CC_CC}
{DDDD}.

5 Box Operations

This section introduces the box operations implemented in `ztool`, including box measurement and simple box transformations.

```
\ztool_get_ht:Nn
\ztool_get_ht:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_ht:Nn <dim>{\<content>}
```

This command is used to save the height of `<content>` in the register `<dim>`.

```
\ztool_get_ht_plus_dp:Nn
\ztool_get_ht_plus_dp:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_ht:Nn <dim>{\<content>}
```

This command is used to save the sum of the height and depth of `<content>` in the register `<dim>`.

```
\ztool_get_wd:Nn
\ztool_get_wd:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_wd:Nn <dim>{\<content>}
```

This command is used to save the width of `<content>` in the register `<dim>`.

```
\ztool_get_dp:Nn
\ztool_get_dp:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_dp:Nn <dim>{\<content>}
```

This command is used to save the depth of `<content>` in the register `<dim>`.

```
\ztool_gget_ht:Nn
\ztool_gget_ht:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_gget_ht:Nn <dim>{\<content>}
```

This command is used to save the height of `<content>` in the register `<dim>`, and this operation is global.

```
\ztool_gget_wd:Nn
\ztool_gget_wd:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_gget_wd:Nn <dim>{\<content>}
```

This command is used to save the width of `<content>` in the register `<dim>`, and this operation is global.

```
\ztool_gget_dp:Nn
\ztool_gget_dp:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_gget_dp:nn <dim>{\<content>}
```

This command is used to save the depth of `<content>` in the register `<dim>`, and this operation is global.

```
\ztool_set_to_wd:nn
\ztool_set_to_wd:(en|ne)
```

Updated: 2024-12-05

```
\ztool_set_to_wd:nn {\<dim>}{\<content>}
```

This command is used to adjust the width of `<content>` to `<dim>`. Its height will be scaled proportionally, and then `<content>` will be typeset. **Note:** This command is different from commands like `\hbox_to_wd:nn`.

```
\ztool_set_to_ht:nn
\ztool_set_to_ht:(en|ne)
```

Updated: 2024-12-05

```
\ztool_set_to_ht:nn {\<dim>}{\<content>}
```

This command is used to adjust the height of `<content>` to `<dim>`. Its width will be scaled proportionally, and then `<content>` will be typeset. **Note:** This command is different from commands like `\vbox_to_ht:nn`.

<code>\ztool_autoset_to_wd_and_ht:nnn</code>	<code>\ztool_autoset_to_wd_and_ht:nn</code>
<code>\ztool_autoset_to_wd_and_ht:(nne een eee)</code>	<code>{\langle width \rangle}{\langle height \rangle}{\langle content \rangle}</code>

Updated: 2025-04-29

This command is used to adjust the width of $\langle content \rangle$ to $\min(\langle width \rangle, \langle height \rangle)$, and then typeset it. See the description of the `\box_autosize_to_wd_and_ht:Nnn` command for details.

<code>\ztool_rotate:nn</code>	<code>\ztool_rotate:nn {\langle angle \rangle}{\langle content \rangle}</code>
<code>\ztool_rotate:(en ne ee)</code>	

New: 2025-04-29

This command is used to rotate $\langle content \rangle$ by $\langle angle \rangle$ degrees and then typeset it.

<code>\ztool_scale_to_wd:nn</code>	<code>\ztool_scale_to_wd:nn {\langle dim \rangle}{\langle content \rangle}</code>
<code>\ztool_scale_to_wd:(en ne ee)</code>	

New: 2025-04-29

This command is used to adjust the width of $\langle content \rangle$ to $\langle dim \rangle$, but without making any adjustments to the height of the box, and then typeset it.

<code>\ztool_scale_to_ht:nn</code>	<code>\ztool_scale_to_ht:nn {\langle dim \rangle}{\langle content \rangle}</code>
<code>\ztool_scale_to_ht:(en ne ee)</code>	

New: 2025-04-29

This command is used to adjust the total height + depth of $\langle content \rangle$ to $\langle dim \rangle$, but without making any adjustments to the width of the box, and then typeset it.

<code>\ztool_scale_to_wd_and_ht:nnn</code>	<code>\ztool_scale_to_wd_and_ht:nnn {\langle width \rangle}{\langle height \rangle}{\langle content \rangle}</code>
<code>\ztool_scale_to_wd_and_ht:(nno nne eee)</code>	

New: 2025-04-29

This command is used to adjust the width of $\langle content \rangle$ to $\langle width \rangle$, and the total height + depth to $\langle height \rangle$, then typeset it.

<code>\ztool_box_item_align:Nnnn</code>	<code>\ztool_box_item_align:Nnnn</code>
<code>\ztool_box_item_align:(cnno Nnno cnno Nnen Nnee)</code>	<code>\langle cmd \rangle{\langle width \rangle}{\langle content \rangle}{\langle align \rangle}</code>

Updated: 2025-05-13

This command is used to adjust the width of $\langle content \rangle$ to $\langle width \rangle$ and then typeset it. $\langle align \rangle$ is used to control the alignment. Optional values are: **left**, **center**, **right**, **scatter**. $\langle cmd \rangle$ is a command that accepts one argument, which will be applied to each token of $\langle content \rangle$. **Note:** Spaces in $\langle content \rangle$ will be ignored. If spaces are needed, please use “ $\backslash_$ ” or “ $\sim$ ” instead.

<code>\ztool_fp_to_rad:n</code>	<code>\ztool_fp_to_rad:n {\langle angle \rangle}</code>
---------------------------------	---

New: 2025-05-12

This command is used to convert $\langle angle \rangle$ from radians to degrees.

<code>\ztoolboxaffine</code>	<code>\ztoolboxaffine[⟨key-value⟩]{⟨content⟩}{⟨matrix⟩}</code>
------------------------------	--

New: 2025-05-12

The above $\langle content \rangle$ represents the object on which the affine transformation acts; $\langle matrix \rangle$ is a 2×2 matrix, representing the corresponding affine transformation matrix. If $\langle matrix \rangle = \{a, b, c, d\}$, its corresponding affine transformation matrix Λ is as follows:

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

If $\det \Lambda = 0$, this transformation is meaningless, `ztool` will output a warning in the terminal, and finally output the content in $\langle content \rangle$ as it is to the PDF; for $\langle keyval \rangle$ settings, please refer to the following “`ztool/affine`” description. **Note:** This command is encapsulated from the `\ztool_affine_transformation:Nnnnn` command below.

<code>ztool/affine/debug</code>	<code>debug</code>	<code>=</code>	$\langle true false \rangle$		Initial value: <code>false</code>
<code>ztool/affine/pole-1</code>	<code>pole-1</code>	<code>=</code>	$\langle coffin's pole \rangle$		Initial value: <code>l</code>
<code>ztool/affine/pole-2</code>	<code>pole-2</code>	<code>=</code>	$\langle coffin's pole \rangle$		Initial value: <code>b</code>
<code>ztool/affine/xoffset</code>	<code>xoffset</code>	<code>=</code>	$\langle number \rangle$		Initial value: <code>Opt</code>
<code>ztool/affine/yoffset</code>	<code>yoffset</code>	<code>=</code>	$\langle number \rangle$		Initial value: <code>Opt</code>

$\langle debug \rangle$ is used for debugging. If set to `true`, some intermediate variable information will be output in the PDF for debugging; $\langle xoffset \rangle$ and $\langle yoffset \rangle$ are the horizontal and vertical offsets, with default values of `Opt`; $\langle pole-1 \rangle$ and $\langle pole-2 \rangle$ are used to set the reference point when printing the coffin, and the two must intersect. For detailed usage of the latter four keys, please refer to the description of `l3coffins`.

<code>\ztool_affine_transformation:Nnnnn</code>	<code>\ztool_affine_transformation:Nnnnn</code>
<code>\ztool_affine_transformation:(Neeee cnnnn ceeee)</code>	$\langle coffin \rangle \{ \langle a \rangle \} \{ \langle b \rangle \} \{ \langle c \rangle \} \{ \langle d \rangle \}$

New: 2025-05-12

This command is used to perform arbitrary affine transformation (linear transformation) on $\langle coffin \rangle$. For the specific usage method, please refer to the aforementioned `\ztoolboxaffine` command. The affine transformation matrix Λ corresponding to the above parameters is

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

Some technical details about the aforementioned function `\zttool_affine_transformation:Nnnnn`: Given any affine transformation Λ , let

$$\Lambda = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}.$$

We can do the following decomposition (similar to SVD decomposition), let $m = 2x$, then we have:

$$\begin{aligned} \Lambda &= \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & m \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} \cos \omega & -\sin \omega \\ \sin \omega & \cos \omega \end{bmatrix} \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}. \end{aligned} \quad (5.1)$$

We give the following notation:

- $\mathbf{T}_1(\theta)$: rotation matrix, rotating θ angle counterclockwise around the origin;
- $\mathbf{T}_2(x)$: scaling matrix, changing all vectors in the x -axis direction to x times the original;
- $\mathbf{T}_3(y)$: scaling matrix, changing all vectors in the y -axis direction to y times the original;

Then we can consider $\{\mathbf{T}_1(\theta), \mathbf{T}_2(x), \mathbf{T}_3(y)\}$ to be the basis of $A_{2 \times 2}$. So we can write the above equation (5.1) as the following expression:

$$\Lambda = \mathbf{T}_1(\theta) \cdot \mathbf{T}_1(\phi) \cdot \mathbf{T}_2(S_x) \cdot \mathbf{T}_3(S_y) \cdot \mathbf{T}_1(\omega) \cdot \mathbf{T}_2(s_x) \cdot \mathbf{T}_3(s_y). \quad (5.2)$$

Based on the result of matrix multiplication, we can know the parameters m, s_x, S_x, ϕ , etc. as follows:

$$s_x = \sqrt{A_{11}^2 + A_{21}^2}, \quad \theta = \arctan\left(\frac{A_{21}}{A_{11}}\right).$$

The results for s_y and m are as follows:

$$ms_y = A_{12} \cos \theta + A_{22} \sin \theta, \quad s_y = \begin{cases} \frac{ms_y \cos \theta - A_{12}}{\sin \theta} & \text{if } \sin \theta \neq 0, \\ \frac{A_{22} - ms_y \sin \theta}{\cos \theta} & \text{if } \sin \theta = 0; \end{cases}$$

Then it is easy to know $m = ms_y/s_y$. The decomposition result for the shear matrix is as follows:

$$\begin{aligned} S_x &= \sqrt{\frac{m^2}{4} + 1} - \frac{m}{2}, & S_y &= \sqrt{\frac{m^2}{4} + 1} + \frac{m}{2}, \\ \phi &= -\frac{\pi}{4} - \frac{1}{2} \arctan\left(\frac{m}{2}\right), & \omega &= \frac{\pi}{4} - \frac{1}{2} \arctan\left(\frac{m}{2}\right). \end{aligned}$$

Finally, we only need to apply this series of transformations to the `\box` from right to left. As can be seen from the above, the command `\zttool_affine_transformation:Nnnnn` only depends on the two functions `\coffin_scale:Nnn` and `\coffin_rotate:Nn` in `LATEX3`.

The relevant reference links during the implementation of the command `\ztool_affine_transformation:Nnnnn` are as follows:

- <https://math.stackexchange.com/a/3521141/1235323>;
- <https://math.stackexchange.com/a/281087/1235323>.

If the original $\text{\TeX}$ engine provided primitives related to shear transformation, then the above decomposition of the shear matrix would be unnecessary. Some engines provide the affine transformation matrix primitive natively, such as the `\pdfsetmatrix` command in $\text{\pdfTeX}$.

The following example shows how to use several box operation commands in this chapter:

Ex. 5

```

\ExplSyntaxOn
\setlength{\fboxsep}{0pt}
% get dim of content
\dotfill\par
\dim_new:N \l_ztool_tmp_H_dim
\dim_new:N \l_ztool_tmp_W_dim
\ztool_get_ht:Nn \l_ztool_tmp_H_dim {Hello,~world!}
\ztool_get_wd:Nn \l_ztool_tmp_W_dim {Hello,~world!}
\dim_use:N \l_ztool_tmp_H_dim \quad \dim_use:N \l_ztool_tmp_W_dim\par

% set content to dim
\dotfill\par
Hello,~world|
\ztool_set_to_ht:nn {.5cm} {Hello,~world}|
\ztool_set_to_wd:nn {40pt} {Hello,~world}\par

% scale one dimension
\dotfill\par
\ztool_scale_to_wd:nn {2em}{\fbox{AA}}\par
\ztool_scale_to_wd:nn {2em}{\fbox{AAA}}\par
\ztool_scale_to_wd:nn {2em}{\fbox{AAAAA}}\par
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}\hbox{A}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}\hbox{A}}\hbox{A}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}\hbox{A}}\hbox{A}}\hbox{A}}}\quad
\par

% box item align
\dotfill\par
\def\boxItemCmd#1{\textcolor{blue}{|#1|}}
\underline{
  \ztool_box_item_align:Nnnn \boxItemCmd{15em}{Tom}{Amy}{Jennery}{scatter}

```

```

}\par
\underline{
  \ztool_box_item_align:Nnnn \boxItemCmd{15em}{\{Tom\} \{Amy\} }
\_{{Jennery}}{center}
}\par

% affine transform
\dotfill\par
\hcoffin_set:Nn \l_tmpa_coffin {\rule{2em}{2em}}
\coffin_typeset:Nnnnn \l_tmpa_coffin {1}{b}{0pt}{0pt}
\ztool_affine_transformation:Nnnnn \l_tmpa_coffin {1}{0}{.5}{1}
\coffin_typeset:Nnnnn \l_tmpa_coffin {1}{b}{0pt}{0pt}
\ExplSyntaxOff
.....
7.8402pt 60.87103pt
.....
Hello, world|Hello, world|Hello, world
.....
AA
AAA
AAAAA
A  A  A  A
A  A  A  A
.....
|Tom|      |Amy|      |Jennery|
|Tom|Amy|  ||Jennery|
.....

```

6 zdraw

This section mainly contains some image drawing commands. This series of commands does not depend on the `tikz` package. Their main dependencies are as follows:

- $\text{\LaTeX} 2_{\epsilon}$ built-in `picture` environment;
- `pict2e`: an enhanced version of the $\text{\LaTeX} 2_{\epsilon}$ built-in `picture` environment, providing better drawing functions;
- `bxeepic`: can be used to provide dash line support, currently **not** included.

<code>\zpic</code>	<code>\begin{zpic}[\langle key-value \rangle] \langle draw commands \rangle \end{zpic}</code>
New: 2025-05-13	This environment is defined based on the $\text{\LaTeX} 2_{\epsilon}$ built-in <code>picture</code> environment.

<code>ztool/draw/picture/unit</code>	<code>unit</code>	<code>= \langle length \rangle</code>	Initial value: <code>1cm</code>
<code>ztool/draw/picture/width</code>	<code>width</code>	<code>= \langle float \rangle</code>	Initial value: <code>0</code>
<code>ztool/draw/picture/height</code>	<code>height</code>	<code>= \langle float \rangle</code>	Initial value: <code>0</code>
<code>ztool/draw/picture/depth</code>	<code>depth</code>	<code>= \langle float \rangle</code>	Initial value: <code>0</code>
<code>ztool/draw/picture/xoffset</code>	<code>xoffset</code>	<code>= \langle float \rangle</code>	Initial value: <code>0</code>
<code>ztool/draw/picture/yoffset</code>	<code>yoffset</code>	<code>= \langle float \rangle</code>	Initial value: <code>0</code>

If `\langle width \rangle` adopts the default value, the current image will overlap with the surrounding typeset content.

<code>\put</code>	<code>\put (\langle x, y \rangle) \{ \langle content \rangle \}</code>
New: 2025-05-13	This command is the same as the <code>\put</code> command in the $\text{\LaTeX} 2_{\epsilon}$ built-in <code>picture</code> environment. Note: This command needs to be used in the <code>picture</code> or <code>zpic</code> environment.

<code>\zline</code>	<code>\zline [\langle key-value \rangle] (\langle coor-1 \rangle) (\langle coor-2 \rangle)</code>
New: 2025-05-13	This command is used to draw a line segment from <code>\langle coor-1 \rangle</code> to <code>\langle coor-2 \rangle</code> . <code>\langle key-value \rangle</code> is used to set the properties of the line. For available options, please refer to the following <code>\parent=ztool/draw/picture/line</code> .

<code>ztool/./line/draw</code>	<code>draw</code>	<code>= \langle color \rangle</code>	Initial value: <code>black</code>
<code>ztool/./line/width</code>	<code>width</code>	<code>= \langle length \rangle</code>	Initial value: <code>.4pt</code>
<code>ztool/./line/dash</code>	<code>dash</code>	<code>= \langle true false \rangle</code>	Initial value: <code>false</code>

The above `\langle width \rangle` is used to set the width of the line, `\langle draw \rangle` is used to set the color of the line, and `\langle dash \rangle` is used to set whether the line is a dashed line. **Note:** Currently the `\langle dash \rangle` option has not been adapted and is **in an unavailable state**.

<code>\zvector</code>	<code>\zvector [\langle key-value \rangle] (\langle coor-1 \rangle) (\langle coor-2 \rangle)</code>
New: 2025-05-13	This command is used to draw a vector. The starting point of the vector is <code>\langle coor-1 \rangle</code> and the end point is <code>\langle coor-2 \rangle</code> . <code>\langle key-value \rangle</code> is used to set the appearance attributes of the vector, which is inherited from <code>\parent=ztool/draw/picture/line</code> . For other available options, please refer to the following <code>\parent=ztool/draw/picture/line/ve</code>

ztool/../../vector/>	> = <code><latex pst></code> Initial value: <code>latex</code> This option is used to control the style of the arrow. The default is <code>L^AT_EX</code> style, i.e., <code>\ltxarrows</code>; <code><pst></code>, i.e., PsTricks, corresponds to the <code>\pstarrows</code> command.
<code>\zdraw</code> New: 2025-05-13	<code>\zdraw [<i><key-value></i>](<i><coor-1></i>)...(<i><coor-n></i>);</code> This command will draw a polyline segment from point <i><coor-1></i> to point <i><coor-n></i>. <i><key-value></i> is inherited from <i><parent=ztool/draw/picture/line></i> and can be used to set the properties of the line. For additional available options, please refer to the following <i><parent=ztool/draw/picture/zdraw></i>. Note: The “;” at the end of this command cannot be omitted, otherwise an error will occur.
ztool/../../zdraw/vector ztool/../../zdraw/cycle ztool/../../zdraw/fill ztool/../../zdraw/shift	<code>vector = <false true></code> Initial value: <code>false</code> <code>cycle = <false true></code> Initial value: <code>false</code> <code>fill = <false true color></code> Initial value: <code>false</code> <code>shift = {<float, float>}</code> Initial value: <code>{0, 0}</code> When <i><fill></i> is set to <code>true</code>, <i><cycle></i> will be automatically set to <code>true</code>; <i><vector></i> is used to set whether to replace each sub-segment with a vector. <i><shift></i> represents the offset in the <i>x</i> and <i>y</i> directions respectively. Note: The <code>{}</code> in the <i><shift></i> option cannot be omitted.
<code>\zarc</code> New: 2025-05-13	<code>\zarc[<i><key-value></i>](<i><float, float></i>)</code> This command is used to draw an arc. (<i><float, float></i>) is its center. By default, a $\frac{1}{4}$ arc is drawn. <i><key-value></i> is inherited from <i><parent=ztool/draw/picture/line></i> and can be used to set the properties of the line. For additional available options, please refer to the following <i><parent=ztool/draw/picture/zarc></i>.
ztool/../../zarc/radius ztool/../../zarc/start ztool/../../zarc/end ztool/../../zarc/fill	<code>radius = <float></code> Initial value: <code>.5</code> <code>start = <float></code> Initial value: <code>0</code> <code>end = <float></code> Initial value: <code>90</code> <code>fill = <false true color></code> Initial value: <code>false</code> <i><start></i> rotates counterclockwise until it ends at angle <i><end></i>; <i><radius></i> is the radius of the arc; <i><fill></i> is used to set the filling color of the arc.
<code>\zcircle</code> New: 2025-05-13	<code>\zcircle[<i><key-value></i>](<i><float, float></i>)</code> This command is based on the aforementioned <code>\zarc</code> command. By default, it will draw a complete circle with (<i><float, float></i>) as the center; the <i><key-value></i> is the same as the <i><key-value></i> option in the aforementioned <code>\zarc</code> command.
<code>\zrectangle</code> New: 2025-05-13	<code>\zrectangle[<i><key-value></i>](<i><coor-1></i>)(<i><coor-2></i>)</code> This command is used to draw a rectangle. (<i><coor-1></i>) and (<i><coor-2></i>) are the coordinates of the two endpoints of the rectangle’s diagonal; <i><key-value></i> is inherited from <i><parent=ztool/draw/picture/line></i>. For other <i><key-value></i> options, please refer to the following <i><parent=ztool/draw/picture/zrectangle></i>.

```
ztool/./zrectangle/arc
ztool/./zrectangle/fill
```

```
arc = <float>.....Initial value: 0
fill = <false|true|color>.....Initial value: false
```

`<fill>` is used to set the filling color of the rectangle, and `<arc>` is used to set the radius corresponding to the rounded corners of the rectangle.

```
\zpin
```

```
New: 2025-07-09
```

```
\zpin[<foreground|background>]{<code>}
```

This command is used to add annotations to the current page. The reference point is the bottom right corner of the current page, and the positive direction is to the right and upward.

```
\zpin{%
  \begin{zpic}%
    \zline[width=5pt] (0, 0) (1, 1)
  \end{zpic}
}
```

Ex. 6

To facilitate understanding of the basic usage of the aforementioned drawing commands, several drawing examples are provided here.

Example 1: Basic line segment drawing commands.

Ex. 7

```

\mbox{\vskip2em
\begin{zpic}[unit=2em]
  \zdraw[fill, cycle] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[cycle, shift={2, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[fill, shift={4, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[draw=red, width=1pt, shift={6, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, shift={8, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, cycle, shift={10, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, fill, shift={12, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, cycle, fill, shift={14, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
\end{zpic}

```









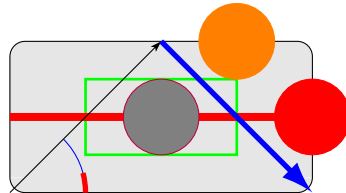

Example 2: Basic geometric element drawing commands.

```

\mbox{\hspace{5em}}
\begin{zpic}[unit=2cm, xoffset=2]
% 1. rectangle
\zrectangle[arc=.1, fill=gray!20](0, 0)(2, 1)
\zrectangle[draw=green, width=1pt](.5, .25)(1.5, .75)
% 2. line / vector
\zline[width=3pt, draw=red](0, .5)(2, .5)
\zvector[>=pst](0, 0)(1, 1)
\zvector[draw=blue, width=2pt](1, 1)(2, 0)
% 3. arc / circle
\zarc[draw=blue, end=45](0, 0) % fill=<empty>
\zarc[draw=blue, width=2pt, end=15, fill=, draw=red](0, 0)
\zcircle[radius=.25, fill, draw=purple](1, .5)
\zcircle[radius=.25, fill=orange, draw=none](1.5, 1)
\zcircle[radius=.25, fill=red, draw=](2, .5)
\end{zpic}

```

Ex. 8



7 TODO

ztool may have changes in the future. Some functions that may be improved in the future are listed here(☐ – unfinished; ☒ – finished; ☒ – functionality not considered):

- ☐ Re-implement the `\xsim_file_write_start:nn` and `\xsim_file_write_stop:` commands in the `xsimverb` package to make them compatible with the `ztool` package.
- ☒ 2025-05-22-Finished: Fixed the issue of the empty first line in the `\ztool_append_to_file:nn` file.
- ☒ 2025-09-01-Finished: For the command `\ztool_read_file_as_seq:nnN`, in some cases all spaces in the source file need to be preserved, reference command `\seq_set_split_keep_spaces:Nnn`.
- ☐ Use the implemented `\ztex_tl_replace_all:nnn` or `\ztex_tl_replace_once:nnn` command to implement `\ztool_replace_file_line_text:nnnn` $\{\langle file \rangle\}\{\langle line \rangle\}$ $\{\langle pattern \rangle\}\{\langle text \rangle\}$, and implement simple regular expression functionality in $\langle pattern \rangle$, ensuring that this command is expandable.
- ☐ Use the `\tl_analysis_map_inline:nn` command from `l3tl-analysis` (which can capture special characters like `'{'`, `'}'`, `'$'`) to implement a simple token debugging command.
- ☐ Use `l3draw` to wrap a `tikz`-like front end, which needs to natively support 3D drawing and automatic occlusion relationship adjustment.
- ☒ 2025-07-09-Finished: Error or unexpected result when `\zline` draws vertical or horizontal line segments.
- ☒ 2026-09-23-Finished: The effects of `\zline` and `\zdraw` are inconsistent. When drawing the same line segment in the same coordinate system, the two cannot coincide (currently it seems the `\zline` command is correct).

8 zTool Source Code

This section lists the source code of the `ztool` package for reference. The line numbers have been specially handled so you can copy the source directly from the PDF.

8.1 ztool.sty

1	%%%	1
2	%% ztool.sty	2
3	%% Copyright 2024, 2025 Zongping Ding.	3
4	%%	4
5	% This work may be distributed and/or modified under the conditions of the	5
6	% LaTeX Project Public License, either version 1.3 of this license or any	6
7	% later version.	7
8	% The latest version of this license is in	8
9	% http://www.latex-project.org/lppl.txt	9
10	% and version 1.3 or later is part of all distributions of LaTeX	10
11	% version 2005/12/01 or later.	11
12	%%	12
13	% This work has the LPPL maintenance status `maintained'.	13
14	%%	14
15	% The Current Maintainer of this work is Zongping Ding.	15
16	%%	16
17	% ztool.sty consists of the parts:	17
18	% shell-escape,	18
19	% file-io,	19
20	% box,	20
21	% zdraw.	21
22	%%%	22
23	\NeedsTeXFormat{LaTeX2e}	23
24	\ProvidesExplPackage{ztool}{2025/05/20}{1.0.1}{A~pre-release~tool~package~for~LaTeX}	24
25		25
26		26
27	%%% 13keys intial patch begin %%%%	27
28	% 1. https://github.com/latex3/latex3/issues/1738	28
29	% 2. https://tex.stackexchange.com/q/742604/294585	29
30	\cs_set_protected:Npn __keys_initialise:n #1	30
31	{	31
32	\exp_after:wN __keys_find_key_module:wNN	32
33	\l_keys_path_str \s_keys_stop	33
34	\l_keys_key_tl \l_keys_key_str	34
35	\tl_set_eq:NN \l_keys_key_tl \l_keys_key_str	35
36	\tl_set:Nn \l_keys_value_tl {#1}	36
37	\cs_if_exist:cTF { \c__keys_code_root_str \l_keys_path_str }	37
38	{	38
39	\str_clear:N \l__keys_inherit_str	39
40	__keys_execute:nn \l_keys_path_str {#1}	40
41	}	41
42	{	42
43	\cs_if_exist:cT	43
44	{ \c__keys_inherit_root_str __keys_parent:o \l_keys_path_str }	44
45	{ __keys_execute_inherit: }	45
46	}	46

47	}	47
48	%%%% 13keys intial patch end %%%%	48
49		49
50		50
51	\clist_new:N \g__ztool_library_loaded_clist	51
52	\clist_gclear:N \g__ztool_library_loaded_clist	52
53	\bool_new:N \g__ztool_lib_user_load_dupuplicate_bool	53
54	\bool_gset_false:N \g__ztool_lib_user_load_dupuplicate_bool	54
55	\cs_new_nopar:Npn __ztool_load_library:n #1	55
56	{	56
57	\clist_map_inline:nn {#1} {	57
58	\clist_if_in:NnTF \g__ztool_library_loaded_clist {##1} {	58
59	\msg_set:nnn {ztool} {library-loaded}	59
60	{	60
61	ztool~library~"##1"~already~loaded,ignored~loading.	61
62	\msg_line_context:	62
63	}	63
64	\bool_if:NT \g__ztool_lib_user_load_dupuplicate_bool	64
65	{	65
66	\msg_warning:nnn {ztool} {library-loaded} {##1}	66
67	}	67
68	}{	68
69	\file_if_exist:nTF {library/ztool.library.##1.tex}{	69
70	\clist_gput_right:Nn \g__ztool_library_loaded_clist {##1}	70
71	\makeatletter\file_input:n {library/ztool.library.##1.tex}	71
72	}{	72
73	\msg_set:nnn {ztool} {library-not-found} {ztool~library~`##1'~not~found.}	73
74	\msg_error:nnn {ztool} {library-not-found} {##1}	74
75	}	75
76	}	76
77	}	77
78	}	78
79	\NewDocumentCommand\ztoolloadlib{m}	79
80	{	80
81	__ztool_load_library:n {#1}	81
82	\bool_gset_true:N \g__ztool_lib_user_load_dupuplicate_bool	82
83	\ExplSyntaxOff	83
84	}	84
85	\keys_define:nn { ztool }	85
86	{	86
87	shell-escape .code:n = { __ztool_load_library:n {shell-escape} },	87
88	file-io .code:n = { __ztool_load_library:n {file-io} },	88
89	box .code:n = { __ztool_load_library:n {box} },	89
90	zdraw .code:n = { __ztool_load_library:n {zdraw} },	90
91	}	91
92	\ProcessKeyOptions [ztool]	92

8.2 shell-escape

```
1 %% 1
2 %% This is file `ztool.library.shell-escape.tex'. 2
3 %% This file is based on the original source code with modifications. 3
4 %% The original disclaimer reads as follows: 4
5 %% 5
6 %% 6
7 %% This is file `l3sys-shell.sty', 7
8 %% generated with the docstrip utility. 8
9 %% 9
10 %% The original source files were: 10
11 %% 11
12 %% l3sys-shell.dtx (with options: `package') 12
13 %% 13
14 %% Copyright (C) 2018,2019 The LaTeX3 Project 14
15 %% 15
16 %% It may be distributed and/or modified under the conditions of 16
17 %% the LaTeX Project Public License (LPPL), either version 1.3c of 17
18 %% this license or (at your option) any later version. The latest 18
19 %% version of this license is in the file: 19
20 %% 20
21 %% http://www.latex-project.org/lppl.txt 21
22 %% 22
23 %% This file is part of the "l3experimental bundle" (The Work in LPPL) 23
24 %% and all files in that bundle must be distributed together. 24
25 %% 25
26 %% File: l3sys-shell.dtx 26
27 \ProvidesExplFile{ztool.library.shell-escape.tex} 27
28 {2025/09/06}{1.0.1} 28
29 {shell-escape-library~for~ztool} 29
30 30
31 31
32 % ==> l3sys-shell tool 32
33 % windows path handle 33
34 \cs_new:Npn \ztool_sys_path_to_win:N #1 34
35 { 35
36 \quark_if_nil:NF #1 { 36
37 \token_if_eq_meaning:NNTF #1 / 37
38 { \c_backslash_str } 38
39 {#1} 39
40 \ztool_sys_path_to_win:N 40
41 } 41
42 } 42
43 \cs_new:Npn \ztool_sys_path_to_win:w #1 ~ #2 \q_stop 43
44 { 44
45 \ztool_sys_path_to_win:N #1 \q_nil 45
46 \tl_if_empty:nF {#2} 46
```

```

47     {
48         \c_space_tl
49         \__sys_path_to_win:w #2 \q_stop
50     }
51 }
52 \cs_new:Npn \ztool_sys_path_to_win:n #1
53 {
54     \exp_after:wN \ztool_sys_path_to_win:w
55     \tl_to_str:n {#1} ~ \q_stop
56 }
57 % respective commands
58 \cs_new_protected:Npn \ztool_shell_escape:n #1
59 {
60     \sys_if_shell_unrestricted:T
61     { \sys_shell_now:n {#1} }
62 }
63 \cs_generate_variant:Nn \ztool_shell_escape:n {e}
64 \cs_new_protected:Npe \ztool_shell_mkdir:n #1
65 {
66     \ztool_shell_escape:e {
67         \sys_if_platform_unix:T
68         {mkdir~\exp_not:N \tl_to_str:n {#1}}
69         \sys_if_platform_windows:T
70         {mkdir~ \exp_not:N \ztool_sys_path_to_win:n {#1}}
71     }
72 }
73 \cs_new_protected:Npe \ztool_shell_cp:nn #1#2
74 {
75     \ztool_shell_escape:e {
76         \sys_if_platform_unix:T
77         {
78             cp~-f~ \exp_not:N \tl_to_str:n {#1} ~
79             \exp_not:N \tl_to_str:n {#2}
80         }
81         \sys_if_platform_windows:T
82         {% can NOT use wildcards in CMD
83             copy~/y~ \exp_not:N \ztool_sys_path_to_win:n {#1} ~
84             \exp_not:N \ztool_sys_path_to_win:n {#2}
85         }
86     }
87 }
88 \cs_new_protected:Npe \ztool_shell_mv:nn #1#2
89 {
90     \ztool_shell_escape:e {
91         \sys_if_platform_unix:T
92         {
93             mv~ \exp_not:N \tl_to_str:n {#1} ~
94             \exp_not:N \tl_to_str:n {#2}

```

```
95     }
96     \sys_if_platform_windows:T
97     {
98         copy~/y~ \exp_not:N \ztool_sys_path_to_win:n {#1} ~
99         \exp_not:N \ztool_sys_path_to_win:n {#2}
100         \token_to_str:N & \token_to_str:N &
101         del~/f~/q~\exp_not:N \ztool_sys_path_to_win:n {#1}
102     }
103 }
104 }
105 \cs_new_protected:Npe \ztool_shell_rm:n #1
106 {
107     \ztool_shell_escape:e {
108         \sys_if_platform_unix:T
109         { rm~-f~ \exp_not:N \tl_to_str:n {#1} }
110         \sys_if_platform_windows:T
111         { del~/f~/q~ \exp_not:N \ztool_sys_path_to_win:n {#1} }
112     }
113 }
114 \cs_new_protected:Npe \ztool_shell_rmdir:n #1
115 {
116     \ztool_shell_mkdir:n {#1}
117     \ztool_shell_escape:e {
118         \sys_if_platform_unix:T
119         { rm~-rf~ \exp_not:N \tl_to_str:n {#1} }
120         \sys_if_platform_windows:T
121         { rmdir~/s~/q~ \exp_not:N \ztool_sys_path_to_win:n {#1} }
122     }
123 }
124 \tl_new:N \l__ztool_shell_tmp_tl
125 \cs_new_protected:Npe \ztool_get_shell_pwd:N #1
126 {
127     \exp_not:N \sys_get_shell:nnN
128     {
129         \sys_if_platform_unix:T { pwd }
130         \sys_if_platform_windows:T { cd }
131     }{
132         \char_set_catcode_other:N \exp_not:N \
133         \char_set_catcode_other:N \exp_not:N \#
134         \char_set_catcode_other:N \exp_not:N \~
135         \char_set_catcode_other:N \exp_not:N \%
136         \char_set_catcode_space:N \exp_not:N \_
137         \tex_endlinechar:D -1 \scan_stop:
138     }
139     \exp_not:N \l__ztool_shell_tmp_tl
140     \str_set:NV #1 \exp_not:N \l__ztool_shell_tmp_tl
141 }
142 \cs_new_protected:Npe \ztool_shell_split_ls:nN #1#2
```

143	{	143
144	\exp_not:N \sys_get_shell:nn	144
145	{	145
146	\sys_if_platform_unix:T { ls~-1~ #1 }	146
147	\sys_if_platform_windows:T { dir~/b~ #1 }	147
148	}{	148
149	\ExplSyntaxOff	149
150	\char_set_catcode_other:N \exp_not:N \	150
151	\char_set_catcode_other:N \exp_not:N \#	151
152	\char_set_catcode_other:N \exp_not:N \~	152
153	\char_set_catcode_other:N \exp_not:N \%	153
154	\char_set_catcode_other:n { 13 }	154
155	}	155
156	\exp_not:N \l__ztool_shell_tmp_tl	156
157	\str_set:NV \exp_not:N \l__sys_tmp_tl \exp_not:N \l__sys_tmp_tl	157
158	\seq_set_split:NnV #2	158
159	{ \char_generate:nn { `^^M } { 12 } }	159
160	\exp_not:N \l__ztool_shell_tmp_tl	160
161	\seq_pop_right:NN #2 \exp_not:N \l__sys_tmp_tl	161
162	}	162
163	\cs_generate_variant:Nn \ztool_shell_mkdir:n {e}	163
164	\cs_generate_variant:Nn \ztool_shell_cp:nn { ee, ne, en }	164
165	\cs_generate_variant:Nn \ztool_shell_mv:nn { ee, ne, en }	165
166	\cs_generate_variant:Nn \ztool_shell_rm:n { e, f, o }	166
167	\cs_generate_variant:Nn \ztool_shell_rmdir:nn { e, f, o }	167
168	\cs_generate_variant:Nn \ztool_get_shell_pwd:N {c}	168
169	\cs_generate_variant:Nn \ztool_shell_split_ls:nN {nc}	169

8.3 file-io

```
1 \ProvidesExplFile{ztool.library.file-io.tex}
2 {2025/09/12}{1.0.1}
3 {file-io~library~for~ztool}
4
5
6 % ==> file IO operations
7 % 1. create a new file
8 % 2. append to a file
9 % 3. read from file / write to file
10 \ior_new:N \g_ztool_file_read_ior
11 \ior_new:N \g_ztool_file_append_ior
12 \iow_new:N \g_ztool_file_append_iow
13 \tl_new:N \l_ztool_current_line
14 \str_clear:N \l_ztool_file_ori_content_str
15 \seq_new:N \l_ztool_file_seq
16 \seq_new:N \l_ztool_tmp_seq
17 \cs_generate_variant:Nn \seq_use:Nn { Ne }
18
19 % read file as seq(not keep internal spaces):
20 \cs_new_protected:Npn \ztool_read_file_as_seq:nnN #1#2#3
21 {% #1: bool(True to keep spaces, False to trim); #2: file name; #3: seq
22   \seq_clear:N #3
23   \file_if_exist:nT {#2}
24   {
25     \ior_open:Nn \g_ztool_file_read_ior {#2}
26     \ior_map_inline:Nn \g_ztool_file_read_ior
27     {
28       \tl_if_empty:nF {##1}
29       {
30         \bool_if:nTF {#1}
31         { \seq_put_right:Nn #3 {##1} }
32         {
33           \seq_put_right:Ne #3
34           { \tl_trim_spaces:n {##1} }
35         }
36       }
37     }
38     \ior_close:N \g_ztool_file_read_ior
39   }
40 }
41 \cs_new_protected:Npn \ztool_gread_file_as_seq:nnN #1#2#3
42 {% #1: bool(True to keep spaces, False to trim); #2: file name; #3: seq
43   \seq_gclear:N #3
44   \file_if_exist:nT {#2}
45   {
46     \ior_open:Nn \g_ztool_file_read_ior {#2}
```

47	\ior_map_inline:Nn \g_ztool_file_read_ior	47
48	{	48
49	\tl_if_empty:nF {##1}	49
50	{	50
51	\bool_if:nTF {#1}	51
52	{ \seq_gput_right:Nn #3 {##1} }	52
53	{	53
54	\seq_gput_right:Ne #3	54
55	{ \tl_trim_spaces:n {##1} }	55
56	}	56
57	}	57
58	}	58
59	\ior_close:N \g_ztool_file_read_ior	59
60	}	60
61	}	61
62	\cs_generate_variant:Nn \ztool_read_file_as_seq:nnN	62
63	{ ne, nnc, nec }	63
64	\cs_generate_variant:Nn \ztool_gread_file_as_seq:nnN	64
65	{ ne, nnc, nec }	65
66		66
67	% read file as seq(keep these internal spaces):	67
68	\tl_new:N \l__ztool_strmap_read_tl	68
69	\cs_new_protected:Npn \ztool_read_file_keep_spaces:nn #1#2	69
70	{	70
71	\file_if_exist:nT {#1}	71
72	{	72
73	\ior_open:Nn \g_ztool_file_read_ior {#1}	73
74	\ior_str_map_inline:Nn \g_ztool_file_read_ior	74
75	{	75
76	\exp_args:Nee \str_if_in:nnF { \tl_head:n {##1} }	76
77	{ \char_generate:nn {37}{12} }	77
78	{	78
79	\tl_set_rescan:Nnn \l__ztool_strmap_read_tl	79
80	{	80
81	\cctab_select:N \c_document_cctab	81
82	\char_set_catcode_space:n { 9 } % tab	82
83	\char_set_catcode_space:n { 32 } % space	83
84	}{ ##1 }	84
85	#2	85
86	}	86
87	}	87
88	\ior_close:N \g_ztool_file_read_ior	88
89	}	89
90	}	90
91	\cs_new_protected:Npn \ztool_read_file_as_seq_keep_spaces:nnN #1#2#3	91
92	{% #1: bool(True to keep trim spaces, False to trim); #2: file name; #3: seq	92
93	\seq_gclear:N #3	93
94	\ztool_read_file_keep_spaces:nn {#2}	94

9595
9696
9797
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9999
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117117
118118
119119
120120
121121
122122
123123
124124
125125
126126
127127
128128
129129
130130
131131
132132
133133
134134
135135
136136
137137
138138
139139
140140
141141
142142

{
 \bool_if:nTF {#1}
 {
 \exp_args:NNo \seq_put_right:Nn #3
 { \l__ztool_strmap_read_tl }
 }{
 \seq_gput_right:Ne #3
 {
 \exp_args:No \tl_trim_spaces:n
 { \l__ztool_strmap_read_tl }
 }
 }
}

\cs_new_protected:Npn \ztool_gread_file_as_seq_keep_spaces:nnN #1#2#3
 {% #1: bool(True to keep trim spaces, False to trim); #2: file name; #3: seq
 \seq_gclear:N #3
 \ztool_read_file_keep_spaces:nn {#2}
 {
 \bool_if:nTF {#1}
 {
 \exp_args:NNo \seq_gput_right:Nn #3
 { \l__ztool_strmap_read_tl }
 }{
 \seq_gput_right:Ne #3
 {
 \exp_args:No \tl_trim_spaces:n
 { \l__ztool_strmap_read_tl }
 }
 }
 }
 }

\cs_generate_variant:Nn \ztool_read_file_as_seq_keep_spaces:nnN
 { ne, nnc, nec }

\cs_generate_variant:Nn \ztool_gread_file_as_seq_keep_spaces:nnN
 { ne, nnc, nec }

% create file / append to file / write to file
\cs_new_protected:Npn \ztool_file_new:nn #1#2
 {% #1: \c_true_bool to allow overwrite; #2: file name
 \bool_if:nT {#1}
 {
 \iow_open:Nn \g_ztool_file_append_iow {#2}
 \iow_close:N \g_ztool_file_append_iow
 }
 }

\cs_new_protected:Npn \ztool_append_to_file:nn #1#2
 {% #1: file name; #2: content

35

```

143 \seq_clear:N \l_ztool_file_seq
144 \file_if_exist:nF {#1}
145 { \ztool_file_new:nn {\c_true_bool}{#1} }
146 \ior_open:Nn \g_ztool_file_append_ior {#1}
147 \ior_str_map_inline:Nn \g_ztool_file_append_ior
148 {
149     \seq_put_right:Nn \l_ztool_file_seq
150     { ##1 }
151 }
152 \iow_open:Nn \g_ztool_file_append_iow {#1}
153 \seq_if_empty:NF \l_ztool_file_seq
154 {
155     \iow_now:Ne \g_ztool_file_append_iow
156     {
157         \seq_use:Ne \l_ztool_file_seq
158         { \iow_newline: }
159     }
160 }
161 \iow_now:Ne \g_ztool_file_append_iow {#2}
162 \iow_close:N \g_ztool_file_append_iow
163 }
164 \cs_generate_variant:Nn \ztool_append_to_file:nn
165 { no, nf, ne, ee }
166
167 \cs_new_protected:Npn \ztool_write_seq_to_file:nNn #1#2#3
168 {% #1:bool; #2:seq; #3:file name
169     \seq_clear:N \l__ztool_tmp_seq
170     \bool_if:nTF { #1 }
171     {
172         \seq_set_eq:NN \l_ztool_file_seq #2
173     }{
174         \ztool_read_file_as_seq:nnN
175         { \c_true_bool }{ #3 }
176         \l__ztool_tmp_seq
177         \seq_concat:NNN \l_ztool_file_seq
178         \l__ztool_tmp_seq #2
179     }
180     \file_if_exist:nF {#3}
181     { \ztool_file_new:nn {\c_true_bool}{#3} }
182     \iow_open:Nn \g_tmpa_iow { #3 }
183     \seq_if_empty:NF \l_ztool_file_seq
184     {
185         \iow_now:Ne \g_tmpa_iow
186         {
187             \seq_use:Ne \l_ztool_file_seq
188             { \iow_newline: }
189         }
190     }

```

191	\iow_close:N \g_tmpa_iow	191
192	}	192
193	\cs_generate_variant:Nn \ztool_write_seq_to_file:nNn	193
194	{ nNe, nNV, nce, ncV }	194
195		195
196	\cs_new_protected:Npn \ztool_replace_file_line:nnn #1#2#3	196
197	{% #1:file name; #2:line index; #3:replacement	197
198	\seq_clear:N \l_ztool_file_seq	198
199	\file_if_exist:nT {#1}{	199
200	\ior_open:Nn \g_ztool_file_read_ior {#1}	200
201	\ior_str_map_inline:Nn \g_ztool_file_read_ior	201
202	{	202
203	\seq_put_right:Nn \l_ztool_file_seq {##1}	203
204	}	204
205	\ior_close:N \g_ztool_file_read_ior	205
206	\seq_set_item:Nnn \l_ztool_file_seq {#2}	206
207	{ #3 }	207
208	\iow_open:Nn \g_ztool_file_append_iow {#1}	208
209	\seq_if_empty:NF \l_ztool_file_seq	209
210	{	210
211	\iow_now:Ne \g_ztool_file_append_iow	211
212	{	212
213	\seq_use:Ne \l_ztool_file_seq	213
214	{ \iow_newline: }	214
215	}	215
216	}	216
217	\iow_close:N \g_ztool_file_append_iow	217
218	}	218
219	}	219
220	\cs_generate_variant:Nn \seq_set_item:Nnn { Nne }	220
221	\cs_generate_variant:Nn \ztool_replace_file_line:nnn	221
222	{ e, ene, eee }	222
223	\cs_new_protected:Npn \ztool_insert_to_file:nnn #1#2#3	223
224	{% #1:file name; #2:line index; #3:content	224
225	\seq_clear:N \l_ztool_file_seq	225
226	\file_if_exist:nT {#1}{	226
227	\ior_open:Nn \g_ztool_file_read_ior {#1}	227
228	\ior_str_map_inline:Nn \g_ztool_file_read_ior	228
229	{	229
230	\seq_put_right:Nn \l_ztool_file_seq {##1}	230
231	}	231
232	\ior_close:N \g_ztool_file_read_ior	232
233	\tl_set:No \l_ztool_current_line	233
234	{ \seq_item:Nn \l_ztool_file_seq {#2} }	234
235	\seq_set_item:Nne \l_ztool_file_seq {#2}	235
236	{ #3\iow_newline:\l_ztool_current_line }	236
237	\iow_open:Nn \g_ztool_file_append_iow {#1}	237
238	\iow_now:Ne \g_ztool_file_append_iow	238

239	{	239
240	\seq_use:Ne \l_ztool_file_seq	240
241	{ \iow_newline: }	241
242	}	242
243	\iow_close:N \g_ztool_file_append_iow	243
244	}	244
245	}	245
246	\cs_generate_variant:Nn \ztool_insert_to_file:nn	246
247	{ ne, nf, ee }	247

8.4 box

```
1 \ProvidesExplFile{ztool.library.box.tex} 1
2 {2026/02/13}{1.0.1} 2
3 {box~library~for~ztool} 3
4 4
5 5
6 % ==> box manipulation tool 6
7 \cs_set:Nn \__ztool_leave_vmode: 7
8 { \ifvmode \leavevmode \fi } 8
9 9
10 \box_new:N \l_ztool_measure_box 10
11 \box_new:N \l_ztool_tmpa_box 11
12 12
13 % catch box dimension 13
14 \cs_new:Npn \ztool_box_set_to:NNn #1#2#3 14
15 { 15
16 \hbox_set:Nn \l_ztool_measure_box {#3} 16
17 \dim_set:Nn #2 {#1 \l_ztool_measure_box} 17
18 \box_set_eq:NN \l_ztool_measure_box \c_empty_box 18
19 } 19
20 \cs_new:Npn \ztool_box_gset_to:NNn #1#2#3 20
21 { 21
22 \hbox_set:Nn \l_ztool_measure_box {#3} 22
23 \dim_gset:Nn #2 {#1 \l_ztool_measure_box} 23
24 \box_set_eq:NN \l_ztool_measure_box \c_empty_box 24
25 } 25
26 \cs_new:Npn \ztool_get_ht:Nn 26
27 { \ztool_box_set_to:NNn \box_ht:N } 27
28 \cs_new:Npn \ztool_get_ht_plus_dp:Nn 28
29 { \ztool_box_set_to:NNn \box_ht_plus_dp:N } 29
30 \cs_new:Npn \ztool_get_wd:Nn 30
31 { \ztool_box_set_to:NNn \box_wd:N } 31
32 \cs_new:Npn \ztool_get_dp:Nn 32
33 { \ztool_box_set_to:NNn \box_dp:N } 33
34 \cs_new:Npn \ztool_gget_ht:Nn 34
35 { \ztool_box_gset_to:NNn \box_ht:N } 35
36 \cs_new:Npn \ztool_gget_wd:Nn 36
37 { \ztool_box_gset_to:NNn \box_wd:N } 37
38 \cs_new:Npn \ztool_gget_dp:Nn 38
39 { \ztool_box_gset_to:NNn \box_dp:N } 39
40 \cs_generate_variant:Nn \ztool_get_ht:Nn 40
41 { Ne, ce } 41
42 \cs_generate_variant:Nn \ztool_get_ht_plus_dp:Nn 42
43 { Ne, ce } 43
44 \cs_generate_variant:Nn \ztool_get_wd:Nn 44
45 { Ne, ce } 45
46 \cs_generate_variant:Nn \ztool_gget_ht:Nn 46
```

47	{ Ne, ce }	47
48	\cs_generate_variant:Nn \ztool_gget_wd:Nn	48
49	{ Ne, ce }	49
50		50
51		51
52	%% modify box content	52
53	% 1. auto scale and rotate (smaller of two)	53
54	\cs_new_protected:Npn \ztool_autoset_to_wd_and_ht:nnn #1#2#3	54
55	{% #1:width; #2:height; #3:object	55
56	\hbox_set:Nn \l_ztool_tmpa_box {#3}	56
57	\box_autosize_to_wd_and_ht:Nnn \l_ztool_tmpa_box {#1}{#2}	57
58	__ztool_leave_vmode:	58
59	\box_use:N \l_ztool_tmpa_box	59
60	}	60
61	\cs_new_protected:Npn \ztool_rotate:nn #1#2	61
62	{% #1:angle; #2:object	62
63	\hbox_set:Nn \l_ztool_tmpa_box {#2}	63
64	\box_rotate:Nn \l_ztool_tmpa_box {#1}	64
65	__ztool_leave_vmode:	65
66	\box_use:N \l_ztool_tmpa_box	66
67	}	67
68	\cs_generate_variant:Nn \ztool_rotate:nn	68
69	{ e, ne, ee }	69
70	\cs_generate_variant:Nn \ztool_autoset_to_wd_and_ht:nnn	70
71	{ nne, een, eee }	71
72		72
73	% 2. width/height scale to same time	73
74	% TODO: if '\dim(content) < dim', spread it to 'dim'.	74
75	\cs_new_protected:Npn \ztool_set_to_wd:nn #1#2	75
76	{% #1:width; #2:object	76
77	\hbox_set:Nn \l_ztool_tmpa_box {#2}	77
78	\box_resize_to_wd:Nn \l_ztool_tmpa_box {#1}	78
79	__ztool_leave_vmode:	79
80	\box_use:N \l_ztool_tmpa_box	80
81	}	81
82	\cs_new_protected:Npn \ztool_set_to_ht:nn #1#2	82
83	{% #1:height; #2:object	83
84	\hbox_set:Nn \l_ztool_tmpa_box {#2}	84
85	\box_resize_to_ht:Nn \l_ztool_tmpa_box {#1}	85
86	__ztool_leave_vmode:	86
87	\box_use:N \l_ztool_tmpa_box	87
88	}	88
89	\cs_generate_variant:Nn \ztool_set_to_wd:nn { e, ne, ee }	89
90	\cs_generate_variant:Nn \ztool_set_to_ht:nn { e, ne, ee }	90
91		91
92	% 3. only scale one dimension	92
93	% NOTE: if boxwd{content} <= given dim, no manipulation	93
94	\cs_new_protected:Npn \ztool_scale_to_wd:nn #1#2	94

95	{	95
96	\hbox_set:Nn \l_ztool_tmpa_box {#2}	96
97	\dim_set:Nn \l_tmpa_dim { \box_wd:N \l_ztool_tmpa_box }	97
98	\fp_set:Nn \l_tmpa_fp	98
99	{	99
100	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	100
101	}	101
102	\box_scale:Nnn \l_ztool_tmpa_box {\l_tmpa_fp}{1}	102
103	__ztool_leave_vmode:	103
104	\box_use:N \l_ztool_tmpa_box	104
105	}	105
106	\cs_new_protected:Npn \ztool_scale_to_ht:nn #1#2	106
107	{% take depth into consideration	107
108	\hbox_set:Nn \l_ztool_tmpa_box {#2}	108
109	\dim_set:Nn \l_tmpa_dim { \box_ht_plus_dp:N \l_ztool_tmpa_box }	109
110	\fp_set:Nn \l_tmpa_fp	110
111	{	111
112	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	112
113	}	113
114	\box_scale:Nnn \l_ztool_tmpa_box {1}{\l_tmpa_fp}	114
115	__ztool_leave_vmode:	115
116	\box_use:N \l_ztool_tmpa_box	116
117	}	117
118	\cs_new_protected:Npn \ztool_scale_to_wd_and_ht:nnn #1#2#3	118
119	{% take depth into consideration	119
120	\hbox_set:Nn \l_ztool_tmpa_box {#3}	120
121	\dim_set:Nn \l_tmpa_dim { \box_wd:N \l_ztool_tmpa_box }	121
122	\dim_set:Nn \l_tmpb_dim { \box_ht_plus_dp:N \l_ztool_tmpa_box }	122
123	\fp_set:Nn \l_tmpa_fp	123
124	{	124
125	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	125
126	}	126
127	\fp_set:Nn \l_tmpb_fp	127
128	{	128
129	\fp_eval:n { min(1, \dim_ratio:nn {#2}{\l_tmpb_dim}) }	129
130	}	130
131	\box_scale:Nnn \l_ztool_tmpa_box {\l_tmpa_fp}{\l_tmpb_fp}	131
132	__ztool_leave_vmode:	132
133	\box_use:N \l_ztool_tmpa_box	133
134	}	134
135	\cs_generate_variant:Nn \ztool_scale_to_wd:nn	135
136	{ e, ne, ee }	136
137	\cs_generate_variant:Nn \ztool_scale_to_ht:nn	137
138	{ e, ne, ee }	138
139	\cs_generate_variant:Nn \ztool_scale_to_wd_and_ht:nnn	139
140	{ nne, nno, eee }	140
141		141
142		142

```

143 %% box content align
144 \seq_new:N \l__ztool_boxitem_seq
145 \cs_set_protected:Npn \ztool_box_item_align:Nnnn #1#2#3#4
146 { % #1:cmd, #2:width, #3:object, #4:align format(left, right, scatter, center)
147   \hb@xt@#2{
148     \tl_map_inline:nn {#3}
149     {
150       \seq_put_right:No \l__ztool_boxitem_seq
151       { \exp_not:N #1{##1} }
152     }
153     \str_case:nnF { #4 }
154     {
155       { left }{ \seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
156       { right }{ \hfill\seq_use:Nn \l__ztool_boxitem_seq {} }
157       { scatter }{ \seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
158       { center }{ \hfill\seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
159       { tower }
160       {
161         \edef\seq@count{\seq_count:N \l__ztool_boxitem_seq}
162         \seq_map_indexed_inline:Nn \l__ztool_boxitem_seq
163         { % ##1: index, ##2: content
164           %% Method II: plain
165           \edef\item@width{\dim_eval:n {#2/(\seq@count+1)}}
166           \hskip\item@width\clap{##2}
167           }\hskip\item@width\hss
168         }
169       { custom }
170       {
171         \def\total@width{#2}
172         \def\align@cmd{#1}
173         \def\align@object{#3}
174         \def\align@format{#4}
175         \tl_use:N \l__ztex_boxitem_align_custom_tl
176       }
177       {\relax}
178     }
179     \seq_clear:N \l__ztool_boxitem_seq
180   }
181 \cs_generate_variant:Nn \ztool_box_item_align:Nnnn
182 { c, Nnno, cnno, Nne, Nnee }
183
184
185 %% affine transformation
186 % REF:
187 % 1. https://math.stackexchange.com/a/3521141/1235323
188 % 2. https://math.stackexchange.com/a/281087/1235323
189 \cs_new:Npn \ztool_fp_to_rad:n #1
190 { \fp_eval:n {#1/pi*180} }

```

```

191 \cs_new:Npn \ztool_matrix_det:nnnn #1#2#3#4 191
192 { 192
193     \fp_eval:n { #1*#4 - #2*#3 } 193
194 } 194
195 % (translation) + x-scale + y-scale + rotate 195
196 \fp_new:N \g_affine_precision_fp 196
197 \fp_set:Nn \g_affine_precision_fp {0.0001} 197
198 \fp_new:N \l__affine_@@_a_fp 198
199 \fp_new:N \l__affine_@@_b_fp 199
200 \fp_new:N \l__affine_@@_c_fp 200
201 \fp_new:N \l__affine_@@_d_fp 201
202 \msg_set:nnn { ztool }{affine-det-zero} 202
203 { 203
204     current~determination~of~the~affine~transformation~ 204
205     matrix~equals~to~zero,~give~up~this~transformation 205
206 } 206
207 207
208 \coffin_new:N \l__affine_trans_coffin 208
209 \cs_generate_variant:Nn \coffin_typeset:Nnnnn { Nxxxx } 209
210 \cs_new:Npn \ztool_affine_transformation:Nnnnn #1#2#3#4#5 210
211 { % #1:box; #2:$a_{11}$; #3:$a_{21}$; #4:$a_{12}$; #5:$a_{22}$. 211
212     \fp_compare:nNnT 212
213         { abs(\ztool_matrix_det:nnnn {#2}{#3}{#4}{#5}) } 213
214         < { \g_affine_precision_fp } 214
215         { \prg_map_break:Nn \l__affine_matrix_det_zero 215
216             { \msg_warning:nn { ztool }{affine-det-zero} }} 216
217     \fp_set:Nn \l__affine_@@_a_fp {#2} 217
218     \fp_set:Nn \l__affine_@@_b_fp {#3} 218
219     \fp_set:Nn \l__affine_@@_c_fp {#4} 219
220     \fp_set:Nn \l__affine_@@_d_fp {#5} 220
221     \__box_affine_transform:N #1 221
222     \prg_break_point:Nn \l__affine_matrix_det_zero { } 222
223     \coffin_typeset:Nxxxx \l__affine_trans_coffin 223
224         { \l__ztool_affine_pole_a_tl } 224
225         { \l__ztool_affine_pole_b_tl } 225
226         { \l__ztool_affine_xoffset_dim } 226
227         { \l__ztool_affine_yoffset_dim } 227
228     } 228
229 \cs_generate_variant:Nn \ztool_affine_transformation:Nnnnn 229
230 { Neeee, cnnnn, ceeee } 230
231 \cs_new:Npn \__box_affine_transform:N #1 231
232 { 232
233     % transform debug 233
234     \bool_if:NT \g_ztool_affine_debug_bool 234
235     { 235
236         \noindent\dotfill\[\begin{bmatrix} 236
237             \fp_use:N \l__affine_@@_a_fp & \fp_use:N \l__affine_@@_c_fp\\ 237
238             \fp_use:N \l__affine_@@_b_fp & \fp_use:N \l__affine_@@_d_fp 238

```

```

239 \end{bmatrix}\]
240 }
241 % get affine parameters
242 \__affine_trans_get_sx:
243 \__affine_trans_get_theta:
244 \__affine_trans_get_sy:
245 \__affine_trans_get_Sx:
246 \__affine_trans_get_Sy:
247 \__affine_trans_get_phi:
248 \__affine_trans_get_omega:
249 % start transform box/coffin
250 \coffin_scale:Nnn #1
251 { \l__box_affine_sx_fp }
252 { \l__box_affine_sy_fp }
253 \coffin_rotate:Nn #1
254 { \ztool_fp_to_rad:n {\l__box_affine_omega_fp} }
255 \coffin_scale:Nnn #1
256 { \l__box_affine_Sx_fp }
257 { \l__box_affine_Sy_fp }
258 \coffin_rotate:Nn #1
259 { \ztool_fp_to_rad:n {\l__box_affine_phi_fp} }
260 \coffin_rotate:Nn #1
261 { \ztool_fp_to_rad:n {\l__box_affine_theta_fp} }
262 }
263 \keys_define:nn { ztool / affine }
264 {
265     debug .bool_gset:N = \g_ztool_affine_debug_bool,
266     debug .initial:n = false,
267     debug .default:n = true,
268     pole-1 .tl_set:N = \l__ztool_affine_pole_a_tl,
269     pole-2 .tl_set:N = \l__ztool_affine_pole_b_tl,
270     pole-1 .initial:n = { l },
271     pole-2 .initial:n = { b },
272     xoffset .dim_set:N = \l__ztool_affine_xoffset_dim,
273     yoffset .dim_set:N = \l__ztool_affine_yoffset_dim,
274     xoffset .initial:n = { 0pt },
275     yoffset .initial:n = { 0pt },
276 }
277 \NewDocumentCommand{\ztoolboxaffine}{0{}m>{\SplitList{,}}m}
278 {% #1:key-value; #2:content; #3:matrix.
279     \group_begin:
280         \keys_set:nn { ztool / affine } {#1}
281         \hcoffin_set:Nn \l__affine_trans_coffin {#2}
282         \ztool_affine_transformation:Nnnnn \l__affine_trans_coffin #3
283     \group_end:
284 }
285 % internal affine transform functions
286 \cs_new:Nn \__ztool_affine_debug_fp:N

```

```

287 {
288     \bool_if:NTF \g_ztool_affine_debug_bool
289     { \string #1 % \show #1
290       ~~~\fp_use:N #1\\
291       }{ \relax }
292 }
293 \fp_new:N \l__box_affine_sx_fp
294 \cs_new:Nn \__affine_trans_get_sx:
295 {
296     \fp_set:Nn \l__box_affine_sx_fp
297     { \fp_eval:n {sqrt(\l__affine_@@_a_fp^2 + \l__affine_@@_b_fp^2)} }
298     \__ztool_affine_debug_fp:N \l__box_affine_sx_fp
299 }
300 \fp_new:N \l__box_affine_theta_fp
301 \cs_new:Nn \__affine_trans_get_theta:
302 {
303     \fp_set:Nn \l__box_affine_theta_fp
304     { \fp_eval:n {atan(\l__affine_@@_b_fp/\l__affine_@@_a_fp)} }
305     \__ztool_affine_debug_fp:N \l__box_affine_theta_fp
306 }
307 \fp_new:N \l__box_affine_msy_fp
308 \cs_new:Nn \__affine_trans_get_msy:
309 {
310     \fp_set:Nn \l__box_affine_msy_fp
311     { \fp_eval:n {
312         \l__affine_@@_c_fp*cos(\l__box_affine_theta_fp)
313         +
314         \l__affine_@@_d_fp*sin(\l__box_affine_theta_fp)
315     } }
316     \__ztool_affine_debug_fp:N \l__box_affine_msy_fp
317 }
318 \fp_new:N \l__box_affine_sy_fp
319 \cs_new:Nn \__affine_trans_get_sy:
320 {
321     \__affine_trans_get_msy:
322     \bool_if:nTF
323     {
324         \fp_compare_p:nNn { abs(sin(\l__box_affine_theta_fp)) }
325         < { \c_zero_fp + \g_affine_precision_fp }
326     }{
327         \fp_set:Nn \l__box_affine_sy_fp
328         {
329             ( \l__affine_@@_d_fp - \l__box_affine_msy_fp*sin(\l__box_affine_theta_fp) )
330             / cos(\l__box_affine_theta_fp)
331         }
332     }{
333         \fp_set:Nn \l__box_affine_sy_fp
334         {

```

```

335         ( \l__box_affine_msy_fp*cos(\l__box_affine_theta_fp) - \l__affine_@@_c_fp ) 335
336         / sin(\l__box_affine_theta_fp) 336
337     } 337
338 } 338
339     \__ztool_affine_debug_fp:N \l__box_affine_sy_fp 339
340 } 340
341 \fp_new:N \l__box_affine_m_fp 341
342 \cs_new:Nn \__affine_trans_get_m: 342
343 { 343
344     \fp_set:Nn \l__box_affine_m_fp 344
345     { \l__box_affine_msy_fp / \l__box_affine_sy_fp } 345
346     \__ztool_affine_debug_fp:N \l__box_affine_m_fp 346
347 } 347
348 \fp_new:N \l__box_affine_Sx_fp 348
349 \fp_new:N \l__box_affine_Sy_fp 349
350 \cs_new:Nn \__affine_trans_get_Sx: 350
351 { 351
352     \__affine_trans_get_m: 352
353     \fp_set:Nn \l__box_affine_Sx_fp 353
354     { sqrt(\l__box_affine_m_fp^2/4 + 1) - \l__box_affine_m_fp/2 } 354
355     \__ztool_affine_debug_fp:N \l__box_affine_Sx_fp 355
356 } 356
357 \cs_new:Nn \__affine_trans_get_Sy: 357
358 { 358
359     \fp_set:Nn \l__box_affine_Sy_fp 359
360     { sqrt(\l__box_affine_m_fp^2/4 + 1) + \l__box_affine_m_fp/2 } 360
361     \__ztool_affine_debug_fp:N \l__box_affine_Sy_fp 361
362 } 362
363 \fp_new:N \l__box_affine_phi_fp 363
364 \fp_new:N \l__box_affine_omega_fp 364
365 \cs_new:Nn \__affine_trans_get_phi: 365
366 { 366
367     \fp_set:Nn \l__box_affine_phi_fp 367
368     { -pi/4 - 1/2*atan(\l__box_affine_m_fp/2) } 368
369     \__ztool_affine_debug_fp:N \l__box_affine_phi_fp 369
370 } 370
371 \cs_new:Nn \__affine_trans_get_omega: 371
372 { 372
373     \fp_set:Nn \l__box_affine_omega_fp 373
374     { pi/4 - 1/2*atan(\l__box_affine_m_fp/2) } 374
375     \__ztool_affine_debug_fp:N \l__box_affine_omega_fp 375
376 } 376

```

8.5 zdraw

```
1 \ProvidesExplFile{ztool.library.zdraw.tex}
2 {2026/09/23}{1.1.0}
3 {zdraw~library~for~ztool}
4
5
6 % ==> ztool draw (based on package 'pict2e' and 'picture' env)
7 \RequirePackage{pict2e}
8
9 % operations precise
10 \fp_const:Nn \c__pic_precise_fp { 0.001 }
11
12 % command/env wrapper
13 \cs_new_protected:Npn \_@@_begin_picture:nnnn #1#2#3#4
14 { \begin{picture}
15     (\fp_eval:n {#1}, \fp_eval:n {#2})
16     (\fp_eval:n {-#3}, \fp_eval:n {-#4}) }
17 \cs_new_protected:Npn \_@@_end_picture:
18 { \end{picture} }
19 \cs_new_protected:Npn \_@@_pic_put:nnn #1#2#3
20 { \put(\fp_eval:n {#1}, \fp_eval:n {#2}){ #3 } }
21 \cs_generate_variant:Nn \_@@_begin_picture:nnnn
22 { VVVV, eeee }
23 \cs_generate_variant:Nn \_@@_pic_put:nnn
24 { VVV, een }
25
26 % picture environment alias
27 % width/height/depth are measured in units of `unit` and describe the
28 % dimensions reported to TeX. In particular, `height` is the extent above
29 % the baseline and `depth` is the extent below it. The picture environment
30 % itself always returns zero depth, so zpic collects its result and restores
31 % the requested box dimensions explicitly.
32 \box_new:N \l__pic_output_box
33 \cs_new:Npn \__pic_to_dim:n #1
34 {
35     \fp_to_dim:n
36     { (#1) * \dim_to_fp:n { \l__pic_unit_dim } }
37 }
38 \keys_define:nn { ztool / draw / picture }
39 {
40     unit .dim_set:N = \l__pic_unit_dim,
41     unit .initial:n = { 1cm },
42     width .fp_set:N = \l__pic_width_fp,
43     width .initial:n = 0,
44     height .fp_set:N = \l__pic_height_fp,
45     height .initial:n = 0,
46     depth .fp_set:N = \l__pic_depth_fp,
```

47	depth .initial:n = 0,	47
48	xoffset .fp_set:N = \l__pic_xoffset_fp,	48
49	xoffset .initial:n = 0,	49
50	yoffset .fp_set:N = \l__pic_yoffset_fp,	50
51	yoffset .initial:n = 0,	51
52	% opacity-color .tl_set:N = \l__pic_opacity_color_tl,	52
53	% opacity-color .initial:n = { white },	53
54	}	54
55	\NewDocumentEnvironment{zpic}{0{}}	55
56	{	56
57	\group_begin:	57
58	\keys_set:nn { ztool / draw / picture } {#1}	58
59	\setlength\unitlength{ \l__pic_unit_dim }	59
60	\hbox_set:Nw \l__pic_output_box	60
61	_@@_begin_picture:VVVV	61
62	\l__pic_width_fp \l__pic_height_fp	62
63	\l__pic_xoffset_fp\l__pic_yoffset_fp	63
64	}{	64
65	_@@_end_picture:	65
66	\hbox_set_end:	66
67	\box_set_wd:Nn \l__pic_output_box	67
68	{ __pic_to_dim:n { \l__pic_width_fp } }	68
69	\box_set_ht:Nn \l__pic_output_box	69
70	{ __pic_to_dim:n { \l__pic_height_fp } }	70
71	\box_set_dp:Nn \l__pic_output_box	71
72	{ __pic_to_dim:n { \l__pic_depth_fp } }	72
73	\box_use_drop:N \l__pic_output_box	73
74	\group_end:	74
75	}	75
76		76
77		77
78	% picture commands alias	78
79	\cs_new:Npn __coord_st:n #1	79
80	{ \clist_item:nn {#1}{1} }	80
81	\cs_new:Npn __coord_nd:n #1	81
82	{ \clist_item:nn {#1}{2} }	82
83	\cs_new:Npn __coord_rd:n #1#2	83
84	{ \clist_item:nn {#1}{3} }	84
85	\cs_new:Npn __coord_st_nd:n #1	85
86	{	86
87	{\clist_item:nn {#1}{1}}	87
88	{\clist_item:nn {#1}{2}}	88
89	}	89
90	\cs_new:Npn __coord_st_nd_rd:n #1	90
91	{	91
92	{\clist_item:nn {#1}{1}}	92
93	{\clist_item:nn {#1}{2}}	93
94	{\clist_item:nn {#1}{3}}	94

```

95 }
96 \cs_generate_variant:Nn \__coord_st:n { V, e }
97 \cs_generate_variant:Nn \__coord_nd:n { V, e }
98 \cs_generate_variant:Nn \__coord_rd:n { V, e }
99 \cs_generate_variant:Nn \__coord_st_nd:n { V, e }
100 \cs_generate_variant:Nn \__coord_st_nd_rd:n { V, e }
101
102 \bool_new:N \l__ztool_invalid_color_bool
103 \cs_new:Npn \__color_safe_use:n #1
104 {
105     \__color_if_valid:nT {#1}
106     { \color{#1} }
107 }
108 \prg_new_conditional:Npnn \__color_if_valid:n #1 {p, T, F, TF}
109 {
110     \def\ztool@targer@color{#1}
111     \def\ztool@color@none{none}
112     \bool_if:eTF
113     {
114         \tl_if_empty_p:e {#1} ||
115         \tl_if_eq_p:NN \ztool@color@none \ztool@targer@color
116     }{ \prg_return_false: }
117     { \prg_return_true: }
118 }
119 \prg_generate_conditional_variant:Nnn \__color_if_valid:n
120 { V, e }{ p, T, F, TF }
121 \cs_generate_variant:Nn \__color_safe_use:n
122 { V, e }
123
124
125 % --> line/vector
126 \fp_new:N \l__draw_vector_slope_fp
127 \fp_new:N \l__draw_vector_normal_fp
128 \fp_new:N \l__draw_vector_xysep_fp
129 \cs_new:Npn \__@@_pic_line:nnn #1#2#3
130 { % #1:x; #2:y; #3:x-distance NOT the length
131     \line(\fp_eval:n {#1}, \fp_eval:n {#2})
132     { \fp_eval:n {#3} }
133 }
134 \cs_new:Npn \__@@_pic_vector:nnn #1#2#3
135 { % #1:x; #2:y; #3:x-distance NOT the length
136     \vector(\fp_eval:n {#1}, \fp_eval:n {#2})
137     { \fp_eval:n {#3} }
138 }
139 \keys_define:nn { ztool / draw / picture / line }
140 {
141     draw .tl_set:N = \l__pic_line_draw_color_tl,
142     draw .initial:n = { black },

```

```

143 % color .meta:n = { draw = #1 }, % alias for 'draw'
144 width .dim_set:N = \l__pic_line_width_dim,
145 width .initial:n = { .4pt },
146 dash .bool_set:N = \l__pic_line_dash_bool,
147 dash .initial:n = { false },
148 }
149 \cs_new_protected:Nn \__pic_set_line_width:
150 {
151     \linethickness{ \l__pic_line_width_dim }
152 }
153 \cs_new_protected:Nn \__pic_set_line_color:
154 {
155     \__color_safe_use:V \l__pic_line_draw_color_tl
156 }
157 \cs_new_protected:Nn \__pic_set_fill_color:
158 {
159     \__color_safe_use:V \l__pic_region_fill_color_tl
160 }
161 \def\z@pic@vector@style{\ltxarrows}
162 \keys_define:nn { ztool / draw / picture }
163 {
164     vector .inherit:n = { ztool/draw/picture/line },
165 }
166 \keys_define:nn { ztool / draw / picture / vector }
167 {
168     > .choice:,
169     > / latex .code:n = {\def\z@pic@vector@style{\ltxarrows}},
170     > / pst .code:n = {\def\z@pic@vector@style{\pstarrows}},
171     > / unknown .code:n =
172     {
173         \msg_set:nnn { ztool }{unknown-arrow-style}
174         { Unknown~arrow~style,~use~'latex'~or~'pst'. }
175         \msg_error:nn { ztool }{unknown-arrow-style}
176     }
177 }
178 \tl_new:N \l__draw_line_type % 'horizontal', 'vertical', 'normal'
179 \cs_new_protected:Npn \ztool_pic_line_vector:nnnn #1#2#3#4
180 {% #1:line/vector; #2:key-value; #3:start coor; #4:end coor;
181     \group_begin:
182     \keys_set:nn { ztool / draw / picture / #1 }{#2}
183     \tl_set:Nn \l__draw_line_type { normal }
184     \fp_compare:nNnTF { \__coor_st:n {#4} - \__coor_st:n {#3} } > { \c__pic_precise_fp }
185     {
186         \fp_set:Nn \l__draw_vector_slope_fp
187         { (\__coor_nd:n {#4} - \__coor_nd:n {#3})
188         / (\__coor_st:n {#4} - \__coor_st:n {#3}) }
189         \fp_set:Nn \l__draw_vector_xysep_fp
190         { abs(\__coor_st:n {#4} - \__coor_st:n {#3}) }

```

```

} {
% NOTE: we do NOT set slope infinte, just set it to '0'
\fp_set:Nn \l__draw_vector_slope_fp {0}
\fp_set:Nn \l__draw_vector_xysep_fp
{ abs(\__coord:n {#4} - \__coord:n {#3}) }
\tl_set:Nn \l__draw_line_type { vertical }
}
\fp_compare:nNnT { abs(\__coord:n {#4} - \__coord:n {#3}) } < {0.001}
{ \tl_set:Nn \l__draw_line_type { horizontal } }
\z@pic@vector@style
\__pic_set_line_width:
\exp_last_unbraced:Ne \__@@_pic_put:nnn {\__coord_st_nd:n {#3}}
{
\__pic_set_line_color:
\str_case:VnF \l__draw_line_type
{
{vertical}{
\cs:w __@@_pic_#1:nnn\cs_end:
{ \l__draw_vector_slope_fp }
{ 1 }
{ \l__draw_vector_xysep_fp }
}
{horizontal}{
\cs:w __@@_pic_#1:nnn\cs_end:
{ 1 }
{ \l__draw_vector_slope_fp }
{ \l__draw_vector_xysep_fp }
}
{normal}{
\cs:w __@@_pic_#1:nnn\cs_end: {1}
{ \l__draw_vector_slope_fp }
{ \l__draw_vector_xysep_fp }
}
}
}{ \relax }
}
\group_end:
}
\cs_generate_variant:Nn \ztool_pic_line_vector:nnnn {neee, nooo}
\NewDocumentCommand{\zline}{0{}d()d()}
{
\ztool_pic_line_vector:neee {line}{#1}{#2}{#3}
}
\NewDocumentCommand{\zvector}{0{}d()d()}
{
\ztool_pic_line_vector:neee {vector}{#1}{#2}{#3}
}

```

```

239 % --> \zdraw -- similar to \tikz command in tikz
240 % NOTE: these line/vector commands are identical to
241 % 1. \Line (x_1, y_1)(x_2, y_2), \Vector (x_1,y_1)(x_2,y_2)
242 % 2. \polyline(x_1, y_1) ... (x_n, y_n), \polyvector(x_1, y_1) ... (x_n, y_n)
243 % 3. \polygon (x_1, y_1) ... (x_n, y_n), when set 'cycle',
244 % \polygon*(x_1, y_1) ... (x_n, y_n), when set 'fill' (auto cycle).
245 % 4. Trim leading space after '\polygon' or '*' to avoid error !!
246 \cs_new:Npn \__@@_pic_Line:nnnn #1#2#3#4
247 { \Line (#1, #2)(#3, #4) }
248 \cs_new:Npn \__@@_pic_Vector:nnnn #1#2#3#4
249 { \Vector (#1, #2)(#3, #4) }
250 \cs_new:Npn \__@@_pic_polyline:n #1
251 {
252   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
253   \exp_last_unbraced:NV \polyline \l_tmpa_tl
254 }
255 \cs_new:Npn \__@@_pic_polyvector:n #1
256 {
257   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
258   \exp_last_unbraced:NV \polyvector \l_tmpa_tl
259 }
260 \cs_new:Npn \__@@_pic_polygon:nn #1#2
261 {
262   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
263   \tl_set:Nc \l_tmpb_tl {\tl_trim_spaces:e {#2}}
264   \tl_set:Nc \l_tmpa_tl { \l_tmpa_tl\l_tmpb_tl }
265   \exp_last_unbraced:NV \polygon \l_tmpa_tl
266 }
267 \cs_generate_variant:Nn \__@@_pic_polygon:nn { nV, ne }
268 \tl_new:N \l__pic_region_fill_color_tl
269 \bool_new:N \l__pic_region_fill_bool
270 \keys_define:nn { ztool / draw / picture / region }
271 {
272   fill .choices:nn = { true, false }{
273     \use:c { bool_set_ \l_keys_choice_tl :N }
274     \l__pic_region_fill_bool
275   },
276   fill .initial:n = { false },
277   fill .default:n = { true },
278   fill / unknown .code:n = {
279     \tl_if_empty:eF \l_keys_value_tl
280     { \bool_set_true:N \l__pic_region_fill_bool }
281     \tl_set:Nc \l__pic_region_fill_color_tl { \l_keys_value_tl }
282   },
283 }
284 \keys_define:nn { ztool / draw / picture }
285 {
286   zdraw .inherit:n = {

```

```

287         ztool/draw/picture/line,
288         ztool/draw/picture/vector,
289         ztool/draw/picture/region,
290     },
291 }
292 \keys_define:nn { ztool / draw / picture / zdraw }
293 {
294     vector .bool_set:N = \l__pic_draw_vector_bool,
295     vector .initial:n = { false },
296     cycle .bool_set:N = \l__pic_draw_cycle_bool,
297     cycle .initial:n = { false },
298     shift .tl_set:N = \l__pic_draw_shift_tl,
299     shift .initial:n = { 0, 0 },
300 }
301 \cs_new:Npn \__region_fill_color_miss:n #1
302 {
303     \bool_if:eT {
304         \l__pic_region_fill_bool &&
305         \tl_if_empty_p:N \l__pic_region_fill_color_tl
306     } { \tl_set:Nn \l__pic_region_fill_color_tl {#1} }
307 }
308 \cs_new_protected:Npn \ztool_pic_draw:nw #1#2;
309 {% #1:key-value; #2:coors list (use ';' to end scan just like tikz)
310     \group_begin:
311     \keys_set:nn { ztool / draw / picture / zdraw } {#1}
312     \__region_fill_color_miss:n { gray }
313     \edef\coors@first
314     {
315         \exp_last_unbraced:Ne
316         \__coors_list_first:w {\tl_trim_spaces:e {#2}}
317         \scan_stop:
318     }
319     \edef\draw@flag
320     {
321         \tl_map_function:nN {
322             \l__pic_draw_vector_bool
323             \l__pic_draw_cycle_bool
324             \l__pic_region_fill_bool
325         } \int_eval:n
326     }
327     \__@@_pic_put:nnn
328     { \__coor_st:V \coors@first + \__coor_st:V \l__pic_draw_shift_tl }
329     { \__coor_nd:V \coors@first + \__coor_nd:V \l__pic_draw_shift_tl }
330     {
331         \__pic_set_line_width:
332         \__pic_set_line_color:
333         \exp_after:wN \int_case:nnF \exp_after:wN {
334             \exp_after:wN \int_from_bin:n \exp_after:wN

```

```

335         { \draw@flag }
336     }{
337         {0}{ \__@@_pic_polyline:n {#2} }
338         {1}{ \__@@_pic_polygon:nn {*}{#2} }
339         {2}{ \__@@_pic_polygon:ne { }{#2} }
340         {3}{ \__@@_pic_polygon:nn {*}{#2} }
341         {4}{ \__@@_pic_polyvector:n {#2} }
342         {5}{
343             \__pic_set_fill_color:
344             \__@@_pic_polygon:nn {*}{#2}
345             \__pic_set_line_color:
346             \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)}
347         }
348         {6}{ \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)} }
349         {7}{
350             \__pic_set_fill_color:
351             \__@@_pic_polygon:nn {*}{#2}
352             \__pic_set_line_color:
353             \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)}
354         }
355     }{\relax}
356 }
357 \group_end:
358 }
359 \cs_new:Npn \__coors_list_first:w (#1)#2\scan_stop:
360 { #1 }
361 \NewDocumentCommand{\zdraw}{0{}}
362 { \ztool_pic_draw:nw {#1} }
363
364
365 % --> arc / circle
366 \cs_new:Npn \__@@_pic_arc:nnnn #1#2#3#4
367 { % #1:fill bool; #2:start angle; #3:end angle; #4:radius
368     \arc #1[\fp_eval:n {#2}, \fp_eval:n {#3}]
369     { \fp_eval:n {#4} }
370 }
371 \cs_new:Npn \__@@_pic_circel:nn #1#2
372 { % #1:fill bool; #2:radius
373     \__@@_pic_arc:nnnn {#1}{0}{360}{#2}
374 }
375
376
377 % --> arc rule: similar to arc by it's a rule
378 \cs_new:Npn \__@@_pic_arcrule:nnn #1#2#3
379 {
380     % To be implemented ...
381 }
382

```

```

383                                     383
384 % --> circle                                     384
385 \keys_define:nn { ztool / draw / picture }       385
386 {                                               386
387     arc .inherit:n = {                         387
388         ztool/draw/picture/line,               388
389         ztool/draw/picture/region,             389
390     },                                         390
391 }                                               391
392 \keys_define:nn { ztool / draw / picture / arc } 392
393 {                                               393
394     radius .fp_set:N = \l__pic_arc_radius_fp,   394
395     radius .initial:n = .5,                    395
396     start .fp_set:N = \l__pic_arc_start_fp,     396
397     start .initial:n = 0,                      397
398     end .fp_set:N = \l__pic_arc_end_fp,         398
399     end .initial:n = 90,                       399
400 }                                               400
401 \prg_generate_conditional_variant:Nnn           401
402     \bool_if:n { e } { p, T, F, TF }           402
403 \cs_new_protected:Npn \ztool_pic_arc:nn #1#2    403
404     {% #1:key-value; #2:coord                  404
405     \group_begin:                             405
406     \keys_set:nn { ztool / draw / picture / arc }{#1} 406
407     \__region_fill_color_miss:n { gray }         407
408     \__color_if_valid:VF \l__pic_region_fill_color_tl 408
409     { \bool_set_false:N \l__pic_region_fill_bool } 409
410     \exp_last_unbraced:Ne \__@@_pic_put:nnn     410
411     { \__coord_st_nd:n {#2} }                  411
412     {                                           412
413         \__pic_set_line_width:                 413
414         \bool_if:eT \l__pic_region_fill_bool   414
415         {                                       415
416             \__pic_set_fill_color:             416
417             \exp_args:Ne \__@@_pic_arc:nnnn {*} 417
418             { \fp_use:N \l__pic_arc_start_fp } 418
419             { \fp_use:N \l__pic_arc_end_fp }   419
420             { \fp_use:N \l__pic_arc_radius_fp } 420
421         }                                       421
422         % NOTE: border must draw after the fill 422
423         \__pic_set_line_color:                 423
424         \exp_args:Ne \__@@_pic_arc:nnnn {}     424
425         { \fp_use:N \l__pic_arc_start_fp }     425
426         { \fp_use:N \l__pic_arc_end_fp }       426
427         { \fp_use:N \l__pic_arc_radius_fp }    427
428     }                                           428
429     \group_end:                               429
430 }                                               430

```

```

431 \NewDocumentCommand{\zarc}{0{}d{}}
432   {% #1:key-value; #2:coord
433     \ztool_pic_arc:nn {#1}{#2}
434   }
435 \NewDocumentCommand{\zcircle}{0{}d{}}
436   {
437     \ztool_pic_arc:nn {start=0, end=360, #1}{#2}
438   }
439
440
441 % --> oval / rectangle
442 % \oval[arc](full-x-width, full-y-width)[part]
443 % part: (l, r) x (t, b)
444 \cs_new:Npn \__@@_pic_oval:nnnn #1#2#3#4
445   {% #1:arc; #2:part; #3:x-width; #4:y-width;
446     \oval
447       [\fp_eval:n {#1}]
448       (\fp_eval:n {#3}, \fp_eval:n {#4})
449       [ #2 ]
450   }
451 \keys_define:nn { ztool / draw / picture }
452   {
453     rectangle .inherit:n = {
454       ztool/draw/picture/line,
455       ztool/draw/picture/region,
456     },
457   }
458 \keys_define:nn { ztool / draw / picture / rectangle }
459   {
460     arc .fp_set:N = \l__pic_rec_arc_fp,
461     arc .initial:n = 0,
462   }
463
464 \int_new:N \l__pic_rec_quadrant_index_int
465 \clist_const:Nn \l__pic_rec_quadrant_shift_prop
466   { {(0, 0)}, {(-\rec@arc, 0)}, {(-\rec@arc, -\rec@arc)}, {(0, -\rec@arc)} }
467 \cs_new_protected:Npn \ztool_pic_rectangle:nnn #1#2#3
468   {% #1:key-value; #2:start coord; #3:end coord;
469     \group_begin:
470     \keys_set:nn { ztool / draw / picture / rectangle }{ fill=false }
471     \keys_set:nn { ztool / draw / picture / rectangle }{ #1 }
472     \edef\rec@width { \fp_eval:n {\__coord_st:n {#3} - \__coord_st:n {#2}} }
473     \edef\rec@height{ \fp_eval:n {\__coord_nd:n {#3} - \__coord_nd:n {#2}} }
474     % check if arc radius too large.
475     \edef\rec@arc
476       {
477         \fp_eval:n { min(min(\rec@height/2, \rec@width/2), \l__pic_rec_arc_fp) }
478       }

```

```

479 \__region_fill_color_miss:n { gray }
480 \__color_if_valid:VF \l__pic_region_fill_color_tl
481 {
482     \bool_set_false:N \l__pic_region_fill_bool
483     \prg_map_break:Nn \l__ztool_pic_rec_fill {}
484 }
485 %% begin fill rounded rectangle
486 \__pic_rectangle_rec_comp_fill:n { #2 }
487 \__pic_rectangle_arc_comp_fill:n { #2 }
488 %% end fill rounded rectangle
489 \prg_break_point:Nn \l__ztool_pic_rec_fill { }
490 \__@@_pic_put:nnn
491 { \__coord_st:n {#2}+\rec@width/2 }
492 { \__coord_nd:n {#2}+\rec@height/2 }
493 {
494     \__pic_set_line_color:
495     \__pic_set_line_width:
496     \__@@_pic_oval:nnnn
497     { \rec@arc }{ }
498     { \rec@width }
499     { \rec@height }
500 }
501 \group_end:
502 }
503
504 % aux function for rectangle area filling in rectangle
505 \cs_new:Npn \__pic_rectangle_rec_comp_fill:n #1
506 {
507     \__@@_pic_put:nnn {\__coord_st:n {#1}+\rec@arc}{\__coord_nd:n {#1}}
508     {
509         \__pic_set_fill_color:
510         \rule
511             {\fp_eval:n {(\rec@width-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
512             {\fp_eval:n {\rec@height*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
513     }
514     \fp_compare:nNnT { \rec@height-2*\rec@arc } > { \c__pic_precise_fp }
515     {
516         \__@@_pic_put:nnn {\__coord_st:n {#1}}{\__coord_nd:n {#1}+\rec@arc}
517         {
518             \__pic_set_fill_color:
519             \rule
520                 {\fp_eval:n {\rec@arc*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
521                 {\fp_eval:n {(\rec@height-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
522         }
523         \__@@_pic_put:nnn {\__coord_st:n {#1}+\rec@width-\rec@arc}{\__coord_nd:n {#1}+\rec@arc}
524         {
525             \__pic_set_fill_color:
526             \rule

```

```

527         {\fp_eval:n {\rec@arc*\dim_to_decimal:n {\l__pic_unit_dim}}pt} 527
528         {\fp_eval:n {(\rec@height-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt} 528
529     } 529
530 } 530
531 } 531
532 % aux function for arc area filling in rectangle 532
533 \cs_new:Npn \__pic_rectangle_arc_comp_fill:n #1 533
534 { 534
535     \int_set:Nn \l__pic_rec_quadrant_index_int { 0 } 535
536     \tl_map_inline:nn 536
537     { 537
538         {\__coord_st:n {#1}+\rec@width-\rec@arc, \__coord_nd:n {#1}+\rec@height-\rec@arc} 538
539         {\__coord_st:n {#1}+\rec@arc, \__coord_nd:n {#1}+\rec@height-\rec@arc} 539
540         {\__coord_st:n {#1}+\rec@arc, \__coord_nd:n {#1}+\rec@arc} 540
541         {\__coord_st:n {#1}+\rec@width-\rec@arc, \__coord_nd:n {#1}+\rec@arc} 541
542     }{ 542
543     \int_incr:N \l__pic_rec_quadrant_index_int 543
544     \edef\qu@drant@index{\int_use:N \l__pic_rec_quadrant_index_int} 544
545     \exp_last_unbraced:Ne \__@@_pic_put:nnn 545
546     { \__coord_st_nd:n {##1} } 546
547     { 547
548         \__pic_set_fill_color: 548
549         \__@@_pic_arc:nnnn {*} 549
550         { (\qu@drant@index-1)*90 } 550
551         { \qu@drant@index*90 } 551
552         { \rec@arc } 552
553     } 553
554 } 554
555 } 555
556 556
557 \NewDocumentCommand{\zrectangle}{0}{d()d()} 557
558 { 558
559     \ztool_pic_rectangle:nnn { #1 }{#2}{#3} 559
560 } 560
561 561
562 562
563 % ==> absolute page coordinate (left, bottom) = (0, 0) 563
564 \NewDocumentCommand{\zpin}{0{background}m} 564
565 { 565
566     \hook_gput_next_code:nn {shipout/#1} 566
567     { 567
568         \put(0pt, -\paperheight) 568
569         { \makebox(0, 0)[bl]{#2} } 569
570     } 570
571 } 571

```

9 Index

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